



Functionalism, Behaviorism, and Mental Testing

IN THE EARLY DECADES OF THE 20TH CENTURY, WUNDT'S STUDENTS AND the indigenous pioneers of American psychology continued to promote and develop their varied conceptions of scientific psychology. Cattell claimed in his 1895 presidential address to the APA that the "wide range of individual interests" of American psychologists demonstrated their "adjustment in a complex environment" (1896, p. 134). Yet these different interests also proved to be forces of division in the following decades, sometimes characterized as the period of the competing "schools" of psychology (Murchison, 1930; Woodworth, 1931).

Titchener's form of structural psychology remained an intellectual force in the first two decades of the 20th century, but became increasingly isolated at Cornell and was eventually displaced by functionalist and behaviorist psychology. As the 20th century developed, psychology in America distanced itself from its philosophical roots, including the notion that scientific psychology should be grounded in the introspective analysis of consciousness. Many psychologists had abandoned introspective methods by the time John B. Watson issued his behaviorist "manifesto" in 1913 (Watson, 1913a). This was partly because of the "imageless thought" debate, but also because they had little use for introspection in their applied educational, industrial, and clinical work. Psychology in America also distanced itself from its roots in German physiological psychology. Although courses and demonstration practicals in laboratory methods continued to be employed in the PhD certification of American psychologists as bona fide practitioners of scientific psychology, the "brass instruments" of the new psychology were increasingly appropriated and adapted by educational, industrial, and clinical psychologists as part of their battery of mental and physical aptitude tests.

Many psychologists who came to embrace Watson's (1913a) call for a behavioral science of prediction and control did so because it suited their already well developed applied interests, not because they were convinced by Watson's arguments or rhetoric. These interests were frequently a product of the various institutional and social pressures and opportunities that promoted the development of

applied psychology in the first two decades of the 20th century. Indeed, Watson's own goal of developing a behaviorist psychology had as much to do with his own professional career interests and institutional realities as his avowed arguments against introspection and in favor of a positivistic science of behavior.

THE TURN TO APPLIED PSYCHOLOGY

There were a variety of reasons why early-20th-century American psychologists turned to applied psychology, beyond their own personal interests and inclinations. One simple reason, familiar to contemporary students of psychology, was the shortage of academic jobs in teaching and research that were available to newly minted PhDs. By the end of the 19th and early 20th centuries, the supply of PhD certified psychologists exceeded the demand of academic departments of psychology, and new graduates were obliged to seek positions in education, industry, and clinical work if they wished to continue to have a career in psychology. Most of Hall's Clark graduates—including most of those in animal psychology—secured positions in education, as professors of pedagogy, educational testers, or heads of normal schools. Henry Herbert Goddard (1866–1957), a student of Hall's, was one of the first psychologists appointed to a full-time nonacademic position: He became director of psychological research at the Vineland Training School for Feeble-minded Girls and Boys in New Jersey in 1906 (O'Donnell, 1985).

Another reason was the public and professional demand for psychological services, evidenced by the early positive public reception of phrenology and mesmerism and stimulated by the applied rhetoric of the early promoters of scientific psychology, such as James, Münsterberg, Hall, and Cattell. In the 1850s railroad companies had considered using phrenologists to select trainmen in the hope of reducing accidents. In 1896 William Lowe Bryan (1860–1955), a student of Hall's, who founded the laboratory and psychology program at the University of Indiana, was commissioned by the Union and Wabash Railway to study habit formation in telegraphy (Bryan & Harter, 1897). In the early 20th century Münsterberg took on the problem of selecting competent trainmen when he was approached by urban electric railway companies seeking to avoid costly lawsuits (O'Donnell, 1985). The early decades of the 20th century witnessed an ever-increasing demand for mental tests in education and industry.

A third reason was the flight from teaching overload. Assistant professors and instructors in psychology (who were often situated in philosophy departments) were required to teach a wide variety of courses. For example, Raymond Dodge (1871–1942), who later became an applied psychologist at Yale, was required to teach courses in psychology, logic, history of philosophy, ethics, history of English literature, pedagogy, and aesthetics at Ursinus College in Pennsylvania (O'Donnell, 1985). Harry Levi Hollingworth (1880–1956), who taught

every available course at Columbia, Teachers College, and Barnard in order to feed himself, claimed that he “became an applied psychologist in order to earn a living” (cited in O’Donnell, 1985, p. 225). James R. Angell recalled how he had to supplement his meager salary “by teaching in the summer, by teaching university extension courses, by lecturing before clubs, and by teaching at local institutions in the late afternoons, at night, or on Saturdays” (Angell, 1936, p. 15, cited in O’Donnell, 1985, p. 223). Watson’s own formulation of behaviorism was motivated by his need to select a popular theme that could be the focus of a textbook written “largely for money.” As Watson admitted to his friend Robert Yerkes (1876–1956), “I am in debt and I’ve got to get out” (letter to Yerkes, 1909, cited in O’Donnell, 1985, p. 226).

A final reason was the growth of state-supported public universities in the West, whose development differed in significant respects from the private universities and liberal arts colleges in the East where psychology first flourished (Bolles, 1983). At western public universities, academic concerns about curriculum and research were subordinated to more practical ideals of efficiency, expertise, and service (Rudolph, 1962). By the end of the first decade of the 20th century, more than a third of American psychologists were employed in these universities, most of them engaged in some form of applied (and usually educational) psychology. Not surprisingly, many of these, such as Max Meyer at the University of Missouri, Joseph Jastrow at the University of Wisconsin, and Paul Weiss (1879–1931) at Ohio State University, were early anticipators and supporters of Watson’s behaviorism (O’Donnell, 1985).

FUNCTIONAL PSYCHOLOGY

The “school” of **functional psychology**, which was associated with the University of Chicago and Columbia University in the early decades of the 20th century, was unusual in a number of respects. In a very real sense it was the intellectual creation of Titchener, who distinguished between his own form of structural psychology, concerned with the experimental analysis of consciousness, and functional psychology, concerned with the functions of consciousness (Titchener, 1898, 1899). James Rowland Angell and Harvey Carr (1873–1954), the acknowledged leaders of the movement, insisted that functional psychology was not committed to any distinctive theoretical or methodological position. Angell claimed that functional psychology merely aimed to broaden the scope of psychology beyond Titchener’s structuralism (Angell, 1907), and Carr maintained that it was equivalent to American psychology, defined by its dual emphasis on scientific rigor and practical application (Carr, 1925). Although functional psychology was officially defined in contrast to Titchener’s experimental analysis of consciousness, which purported to represent the Wundtian tradition in America, functional psychology was itself remarkably

Wundtian in orientation, given its emphasis on the active, creative, and purposive role of consciousness in the generation and control of adaptive behavior.

Historians of psychology often represent functional psychology as grounded in Darwin's theory of evolution by natural selection, as did functional psychologists themselves (Angell, 1907, 1909). Yet functional psychology owed little to the hereditarian determinism of Darwin's theory. Functional psychologists did not develop systematic explanations of human and animal psychology and behavior in terms of the survival value of inherited psychological and behavioral traits. Although they recognized the survival value of consciousness as a product of evolution, they focused on the conscious and purposive adaptation of individual organisms to their environment during their lifetimes, rather than on the natural selection of psychological traits and behavior in the evolution of human and animal species. Darwin had recognized that man has "great power of adapting his habits to new conditions of life" (1871, p. 158), but insisted that this power was itself a product of natural selection and constrained by inherited instincts (Sohn, 1976). In contrast, functional psychologists laid much greater emphasis on the *plasticity* of human psychology and behavior, to the point that they sometimes came close to denying a significant role to instincts—as did later behaviorists such as Kuo (1921) and Watson (1924/1930). They suggested that humans had evolved to such a degree that they were able to surmount the constraints of natural selection—at least with the aid of scientific psychology. As earlier proponents of the new psychology had seen the experimental analysis of consciousness as an affirmation of the reality and autonomy of consciousness, functional psychologists saw individual adaptation and learning as an affirmation of the reality and efficacy of consciousness and purpose in human behavior and development.

Baldwin and Titchener on Reaction Time

The origins of the distinction between structural and functional psychology can be traced to the dispute between Titchener and James Mark Baldwin (1861–1934) over the measurement of sensory and motor reaction times. Baldwin had been a student of McCosh's at Princeton. He met Cattell when he visited Wundt's laboratory in 1885, and they later founded the journal *Psychological Review* (in 1894). After returning to Princeton to complete his degree, Baldwin accepted the chair of logic and metaphysics at the University of Toronto in 1889, where he set up the first Canadian laboratory the following year. He produced two introductory psychology texts, *Senses and Intellect* in 1889 and *Feeling and Will* in 1891, which might have been very successful had they not been eclipsed by James's *Principles* in 1890. However, they were sufficient to establish his reputation and gain him the Stuart chair in psychology at Princeton, where he set up another laboratory

in 1893 and conducted the series of experimental studies on reaction time that brought him into conflict with Titchener.

Wundt's studies of reaction time employing trained experimental subjects in the Leipzig laboratory had indicated that sensory reaction times are longer than motor reaction times. Titchener had replicated this result with trained experimental subjects in the Cornell laboratory. Baldwin, working with untrained subjects at Princeton, found that motor reaction times are often longer than sensory reaction times and that there are significant individual differences in both sensory and motor reaction times. Baldwin claimed that these differences in reaction times were a product of differences in practice and attention.

The ensuing dispute between Titchener and Baldwin over **type versus practice theories of reaction time** (Baldwin, 1895a, 1896; Titchener, 1895a, 1895b, 1896) was expressive of a fundamental disagreement between them concerning the subject matter and methods of psychology. For Titchener, scientific psychology was directed to the study of the universal dimensions of the normal adult human mind via the introspective reports of trained subjects, based upon the method of controlled experimentation in physics and physiology. Baldwin maintained that scientific psychology was as much concerned with the study of individual differences in human psychology and behavior, based upon the naturalistic observational methods of developmental biology and comparative psychology.

Titchener dismissed Baldwin's results as unscientific because Baldwin employed untrained subjects. Baldwin responded that Titchener's results were experimental artifacts and criticized as objectionably circular Titchener's treatment of a subject's ability to attain anticipated results as a criterion of introspective competence. Titchener was incensed and consequently refused to publish in *Psychological Review* while Baldwin edited it.

Yet by the turn of the century most psychologists had come to reject Wundt and Titchener's view that experiments in psychology must employ trained psychologists. As Cattell put it,

It is usually no more necessary for the subject to be a psychologist than it is for the vivisected frog to be a physiologist.

—(1904, p. 180)

Individual Differences Baldwin's claim that individual differences in human psychology and behavior were a legitimate subject of psychological study was repeated by later functional psychologists. Since Darwin's theory of evolution by natural selection was based upon postulated individual differences in inherited characteristics, functional psychology is often represented as a development of Darwin's theory, on a par with Galton's (1889) anthropometric studies of individual differences.

Yet although Baldwin and later functional psychologists made variability the key to understanding evolution and development, they focused on the variability of the *conscious and purposive adaptation of individual organisms to their environment*, rather than the random variation of inherited characteristics subject to mechanistic natural selection that was the focus of Darwin's theory of evolution and Galton's anthropometric studies.

Baldwin acknowledged the existence of evolved human (and animal) instincts, but championed the role of plasticity in the conscious and purposive adaptation of individuals to the environment. He claimed that "social heredity" through imitation was the source of distinctive human capacities such as language and morality. He consequently laid great stress on the social development of mentality, consciousness, and behavior, which he documented in *Mental Development in the Child and the Race* in 1895, *Social and Ethical Interpretations in Mental Development* in 1897, and *The Individual and Society* in 1911, works that anticipated the later social developmental theories of Charles Horton Cooley (1864–1929) and George Herbert Mead (1863–1931). While he recognized that consciousness, plasticity, and gregariousness (the basis of imitation) were themselves products of evolution, Baldwin claimed that intelligent and purposive adaptation had come to displace instinct and heredity in the higher animals:

One of the most striking features . . . of the evolution of mammals is the progress made by the brain. It is the organ of increasing plasticity and "educability." Its evolution has been correlated with the decline of the instinctive and completely congenital functions. As we advance upward in the mammalian scale, we find decreasing instinctive endowment and increasing plasticity, accompanied by increasing mental capacity and educability.

—(1909, p. 23)

He also claimed that "social heredity" had replaced physical heredity in the transmission and development of distinctively human psychology and behavior:

All the resources of "social transmission"—the handing down of intelligent acquisitions by parental instruction, imitation, gregarious life, etc.—come in directly to take the place of the physical inheritance of such adaptations.

—(1902, p. 81)

While he maintained that his theory of "social heredity" was consistent with the theory of evolution by natural selection, Baldwin followed Wallace (1864) in claiming that humans had evolved to such a degree of conscious intelligence that they had freed themselves from the pressures of natural selection and surmounted instinctual constraints on behavior (1902, p. 144). In this fashion Baldwin provided a defense of human freedom against the hereditarian determinism of

Darwin's theory of evolution (O'Donnell, 1985), by claiming that thought and will had emancipated humans from the constraints of natural selection:

Thinking and willing stand for the opposite of that fixity of structure and directness of action which characterize the life of instinct. . . . The intelligence secures the widest possible range of personal adjustments, and in doing so widens the sphere of organic selection, so that *the creature that thinks has a general screen from the action of natural selection*. . . . This means that with the growth of intelligence, creatures free themselves more and more from the direct action of natural selection.

—(1902, p. 145)

Baldwin moved to Johns Hopkins in 1903 to revive the psychology program and laboratory (which had been decimated when Hall took his equipment, colleagues, and students to Clark University). By that time Baldwin had virtually abandoned laboratory work and focused upon the promotion of his genetic psychological theories of development. He had hoped to reorient the program to developmental studies of mental function rather than the experimental analysis of consciousness, but institutional and local community pressures to develop applied forms of psychology such as educational psychology frustrated his ambition. However, he managed to satisfy his own goals and institutional demands for educationally relevant research by persuading the university to establish a substantial animal laboratory. He hoped that animal psychology would provide experimental support for the prevalence of imitative learning.

Baldwin was the editor of the *Dictionary of Philosophy and Psychology* (1901). He founded the journal *Psychological Bulletin* in 1904 and published a *History of Psychology* in 1913. His professional psychological career in America came to an abrupt end when he was dismissed from Johns Hopkins after being discovered in a brothel in 1908. A man of independent means, he spent most of the rest of his life traveling, mainly in South America and Europe (he ended his career teaching at the Sorbonne in Paris). He published his autobiography, *Between Two Wars*, in 1926.

John Dewey: Purpose and Adaptation

Anticipations of the functionalist program can be found in James, Münsterberg, Hall, Cattell, and Baldwin. However, it is generally recognized that the movement that came to be known as functional psychology was developed by John Dewey and James Roland Angell and inaugurated institutionally when they came to the newly founded department of philosophy at the newly founded University of Chicago in 1894. The University of Chicago was a natural home for this distinctively American form of psychology, since it had been created with Rockefeller money to serve the social needs of the city as well as the intellectual needs of the nation.

John Dewey was born in Burlington, Vermont. As an undergraduate at the University of Vermont he became interested in philosophy and pursued his study of the subject independently while working as a high school teacher. He was accepted by Johns Hopkins as a graduate student in philosophy in 1882 (although he failed to win the competitive scholarship, which went to Cattell). While at Hopkins Dewey studied with Hall, but was probably more influenced by the philosopher George S. Morris (1840–1889), a neo-Kantian and neo-Hegelian idealist. After attaining his PhD in philosophy in 1884 (with a dissertation on Kant's philosophy), Dewey accepted a job teaching philosophy and psychology at the University of Michigan, where he remained for 10 years (except for one year at Minnesota). While at Michigan he published *Psychology* (1886a), arguably the first American textbook of scientific psychology. The work was a peculiar mix of Hegel and anticipations of functional psychology, which sold well for a few years until (like many early textbooks of psychology) it was eclipsed by James's *Principles*.

Dewey was invited to become chair of the newly established department of philosophy at the University of Chicago in 1894, which at that time included psychology and pedagogy. Although he became an important educational theorist, Dewey was not himself a great teacher: He lectured in a monotone that encouraged somnolence rather than creative thinking. His major achievement at the University of Chicago was the creation of an experimental (or laboratory) school in which new teaching practices could be explored. This innovation became a stimulus for the progressive movement in education, championed by Dewey himself, who devoted his 1899 presidential address to the APA, "Psychology and Social Practice," to the advocacy of progressive education (Dewey, 1900).

Dewey opposed the forms of rote learning then prevalent in schools and argued for a more flexible approach that exploited creative student learning through practice, which created some conflict with the education department at the University of Chicago. Dewey moved to Teachers College at Columbia University in 1904, where he remained until his retirement in 1930. In later years he focused on issues in philosophy and education, and his contributions to psychology were subsidiary to this end. He was active in the New York Teacher's Union, the American Association of University Professors (which he helped found with Cattell), and the American Civil Liberties Union. After his retirement in 1930, he sold vegetables on Long Island until he died at the age of 92.

Like Baldwin, Dewey laid great stress upon the plasticity of human psychology and behavior:

There must be a constant growth, adjustment to new relations, intellectual and moral, and this requires plasticity, variability.

—(Dewey, 1886b, p. 260)

Also like Baldwin, Dewey claimed that humans are capable of transcending the mechanistic constraints of evolution. Through his emphasis on the role of conscious purpose in adaptive responses to environmental change and his interventionist approach to education, Dewey championed the view that scientific psychology and pedagogy could develop strategies for surmounting human limitations and promoting positive psychological and social change (Dewey, 1900). Dewey's effective denial of instinctual constraints on human psychology and behavior is perhaps best illustrated by his famous claim that (with the exception of certain basic instincts, the subject matter of "biological psychology") "all psychology is . . . social psychology" (Dewey, 1917, p. 276). By this he meant that most psychological and behavioral capacities and liabilities are the product of social learning, which can be redirected through alternative forms of social learning through education and training.

The Reflex Arc Dewey's major psychological contribution to functional psychology was "The Reflex Arc Concept in Psychology" (1896), the paper that Titchener christened as the first explicit statement of the functionalist program. In this paper Dewey attacked the atomism and mechanism of prevalent conceptions of the reflex arc, which he argued ought to be conceived as a circuit rather than an arc. Dewey claimed that adaptive adjustments to the environment cannot be reduced to discrete stimulus-response sequences, any more than consciousness can be reduced to an aggregation of elemental sensational units. Using James's example of a child attracted to a flame, Dewey argued that the original stimulus is transformed by the child's adaptive behavior of withdrawing her hand from the flame when it is burned, so that the flame is consequently perceived as a source of pain rather than attraction (1896, p. 359).

Dewey maintained that psychologists needed to consider the significance of the adaptive response for the organism, since such behavior is always directed "to a given end" (1886b, p. 245). According to Dewey, the mechanistic reduction of adaptive behavior to discrete stimulus-response sequences ignores the purposive direction of behavior, within a continuous process of learning guided by consciousness:

The fact is that stimulus and response are not distinctions of existence, but teleological distinctions, that is distinctions of function, or part played, with reference to reaching or maintaining an end.

—(1896, p. 365)

Dewey's paper was as much an anticipatory critique of the atomism and mechanism of the behaviorist stimulus-response psychology that came to displace functional psychology as a critique of the artificiality and sterility of Titchener's structural psychology. While his complaints about the artificiality of reductive

analysis applied to the experimental analysis of the elements of consciousness, these were acknowledged by both Wundt and Titchener. Ironically, this founding document of functional psychology, with its emphasis on psychic unity, the creative nature of cognition, and the critical role of consciousness in purposive adaptation seems to have been largely inspired by Wundt's voluntaristic psychology (see Shook, 1995, for a detailed defense of this view). Dewey also followed Wundt in maintaining the autonomy of "psychical" explanation with respect to physiological explanation, particularly with respect to what both held to be the irreducibly teleological component of psychological explanation.

Dewey recognized that Darwin's theory of evolution by natural selection had eliminated the extrinsic teleology of a divinely created and fixed natural order, but insisted that the intrinsic teleology of conscious and intelligent adaptive behavior was the primary engine of human psychological and social development. He rejected explanations of the purposiveness of human psychology and behavior in terms of naturally selected inherited characteristics because they made "the teleological an accidental product of the mechanical" (Dewey, 1886b, pp. 249–250). Consequently there was some point to Titchener's complaint that Dewey and later functional psychologists were in danger of returning psychology to the (Aristotelian) teleological conception of psychological functions that Darwin's mechanistic theory of evolution by natural selection had supposedly displaced (1898, p. 453).

Psychologists frequently rank Dewey's paper as one of the most influential in the history of psychology (Leahey, 1992). This is rather surprising, since Dewey's critique of atomism and mechanism seems to have had little impact on the behaviorist movement that came to displace functional psychology (Manicas, 1987).

James Rowland Angell: The Province of Functional Psychology

James Rowland Angell (1869–1949) took over from Dewey as head of the department of psychology at the University of Chicago when Dewey left for Teachers College in 1904 and came to be identified as the leader of the functionalist movement. During his tenure at Chicago he made the department of psychology the recognized center of functional psychology. He trained a generation of psychologists committed to the eclectic functionalist approach and provided a guiding statement of functionalist principles (Angell, 1904, 1907).

Like Dewey, Angell was born in Vermont. His father was president of the University of Vermont and later of the University of Michigan, where Angell studied psychology with Dewey as an undergraduate. He received his degree in 1890, staying on an extra year to complete an MA in philosophy in 1891. Inspired by James's *Principles*, Angell spent a year at Harvard and completed a second MA in psychology in 1892. On the advice of his brother Frank (who had studied at

Leipzig and set up the laboratory at Cornell University), he tried to work with Wundt, but found to his disappointment that the Leipzig program was fully booked. In an effort to gain the coveted German qualification, Angell traveled to Germany in 1892, where he attended some lectures by Ebbinghaus and enrolled in the graduate program at the University of Halle. He never completed his conditionally accepted dissertation on Kant (accepted on condition that he improve upon his German) because he ran out of funds. Angell returned to the United States in 1893 to take up an instructorship in philosophy and psychology at the University of Minnesota (and to get married). Although he later became president of Yale (where he founded the Institute of Human Relations, home to the neobehaviorist program of Clark L. Hull) and received many honorary degrees, Angell never completed his own doctoral degree.

The following year Dewey offered his former student a position at the University of Chicago. During his 25-year tenure at Chicago, Angell forged the PhD program in psychology into a major force in American psychology and produced the closest thing to a manifesto for functional psychology in his 1906 presidential address to the APA, "The Province of Functional Psychology." Angell claimed that functional psychology was not a distinctive school, but more of "a point of view, a program, an ambition" (1907, p. 61). He acknowledged the legitimacy of both the atomistic theory and introspective methodology of structural psychology, objecting only to its restrictive definition of psychology in terms of the experimental analysis of the elements of consciousness.

The Utilities of Consciousness Angell claimed that functional psychology was concerned with mental operations or processes rather than mental elements or contents. According to Angell, the aim of the functional psychologist was to

discern and portray the typical *operations* of consciousness under actual life conditions, as over against the attempt to analyze and describe its elementary and complex *contents*.

—(1907, pp. 62–63)

Angell claimed that the primary focus of functional psychology was on the role of consciousness in the generation and control of adaptive behavior. For Angell, this naturally led the functional psychologist to take greater interest in recent developments in animal psychology, which he characterized as one of "the most pregnant [movements] with which we meet in our own generation," and to explore the possible contributions of functional psychology to applied disciplines such as "pedagogy and mental hygiene (1907, p. 69).

Yet unlike later behaviorists whose program was also in large part inspired by animal psychology, Angell maintained that the goal of functional psychology was to understand the "fundamental utilities" of consciousness, which he believed would contribute significantly to applied forms of psychology. He aligned

functional psychology with the biological theories of Darwin and Spencer (and Aristotle), but extended talk about the utilities of consciousness way beyond the Darwinian notion of a characteristic naturally selected in accord with past survival value to a teleological conception in which conscious purpose played as significant a role in the generation and control of adaptive behavior as it did in Wundt's voluntaristic psychology and *Völkerpsychologie*.

When Angell left the University of Chicago in 1921 to become president of Yale (after a year as president of the Carnegie Corporation), he handed leadership of the functionalist movement over to Harvey A. Carr, along with the chairmanship of the psychology department. Carr maintained essentially the same position as Angell and claimed that functional psychology was so eclectic that it was equivalent to American psychology. He treated other avowed schools of psychology, such as structuralism, psychoanalysis, Gestalt psychology, and behaviorism as merely exaggerated and restrictive developments of one aspect of functional—or American—psychology (Carr, 1925). Carr stressed the importance of the study of adaptive behavior, supported mental testing, and promoted the development of educational, industrial, clinical, and other forms of applied psychology. However, he also defended traditional forms of physiological psychology and wrote a book on space perception in 1935. As in the case of Angell and Dewey, one of the striking features of Carr's statement of the functionalist program was how thoroughly Wundtian it was. He championed a teleological conception of adaptive behavior based upon conscious purpose and control and even embraced a form of *Völkerpsychologie*:

The nature of mind may also be studied indirectly through its creations and products—industrial inventions, literature, art, religious customs and beliefs, ethical systems, political institutions, etc. This method might well be termed the social avenue of approach.

—(1925, p. 10)

Carr acknowledged the scientific objectivity of both introspection and behavioral observation, whether naturalistic or experimental. However, during his tenure at Chicago the research emphasis shifted toward the experimental analysis of behavior, including animal behavior. Carr served as chair of the department of psychology between 1919 and 1938, during which time it awarded around 150 PhD degrees, ensuring that functional psychology had a significant influence on the consequent development of American psychology.

Edward L. Thorndike and Robert Sessions Woodworth (who were both former students of Cattell's) promoted functionalist forms of psychology at Columbia University, although neither identified themselves as functionalist psychologists, and Woodworth strenuously objected to being identified with any school of psychology. Woodworth published *Contemporary Schools of Psychology* in 1931, which

included chapters on structural psychology, functional psychology, behaviorism, Gestalt psychology, and psychoanalysis, but refused to treat any as the definitive school of American psychology:

Every school is good, though no one is good enough. No one of them has the full vision of the psychology of the future.

—(1931, p. 255)

Social Engineering

Functional psychologists believed that they could exploit their knowledge of human adaptability to improve education, training, personnel selection, and mental health and to relieve psychological and behavioral disorders that were products of individual maladjustment. Their pragmatic vision of the goal of psychology echoed James's plea for a science of "practical prediction and control" that would make a significant contribution to the improvement of the human condition. For James, control of human psychology and behavior through scientific psychology promised to be "an achievement compared with which the control of the rest of physical nature would appear comparatively insignificant" (1892, p. 148).

This vision of psychology as a science that could surmount the constraints of evolution by natural selection and improve the human condition was shared by later behaviorist psychologists and by eugenicists in the mental testing movement, who argued that many aspects of human psychology and behavior are determined by heredity. Behaviorists argued that scientific psychologists could and should improve the human condition by modifying human psychology and behavior through the manipulation and control of its environmental determinants. Eugenicists argued that scientific psychologists could and should improve the human condition by artificially selecting desirable psychological and social traits through positive and negative eugenic programs.

In this critical respect, functional psychologists and other early-20th-century American psychologists and social scientists did embrace one critical implication of Darwin's theory of evolution by natural selection: namely, that natural selection does not ensure human progress. Baldwin had developed his own evolutionary and developmental theories to counter Spencer's form of "laissez-faire" social Darwinism (O'Donnell, 1985), according to which evolution should be allowed to continue unhindered by artificial social intervention. Functional psychologists, behaviorist psychologists, and eugenicists were disinclined to let evolution develop unhindered and committed themselves to social interventionist programs based upon scientific psychology that were purposively directed to the improvement of the human condition.

However, it is not clear whether this was a consequence of their commitment to this implication of Darwin's theory or their independent pragmatist principles.

For in this respect functional psychologists, behaviorist psychologists, and eugenicists replicated the missionary zeal of early American moral philosophers, who likewise championed the social benefits of common sense and experimental psychology, but who were firmly committed to the purposive and progressive nature of divinely guided evolution (Fuchs, 2000).

BEHAVIORISM

In *Behaviorism at Fifty* (1963), B. F. Skinner observed that

Behaviorism, with its emphasis on the last syllable, is not the scientific study of behavior but a philosophy of science concerned with the subject-matter and methods of psychology.

—(p. 951)

All behaviorists were committed to the view that observable behavior (as opposed to conscious experience) is the subject matter of scientific psychology, but they differed on a variety of substantive issues.

The development of American behaviorism may be usefully characterized in terms of three distinct but related phases. The first phase, often called **behaviorism**, was the position advocated by John B. Watson. Watson saw psychology as a positivist science restricted to the correlation of observable stimuli and responses and rejected mentalistic explanations of behavior. He was also (at least in his later years) a committed environmentalist who maintained that most human behavior can be modified through the manipulation of environmental conditions. The second phase, often called **neobehaviorism**, was represented by Edward C. Tolman (1886–1959) and Clark L. Hull (1884–1952). Tolman and Hull agreed that observable behavior is the subject matter of scientific psychology, but acknowledged the legitimacy of mentalistic explanations of behavior. The third phase, often called **radical behaviorism**, was represented by B. F. Skinner and his followers. Skinner, like Watson, saw psychology as a form of positivist science restricted to the correlation of observable stimuli and responses and rejected mentalistic explanations of behavior. Also like Watson, Skinner was a committed environmentalist who claimed that most human behavior can be modified via the manipulation of environmental conditions (contingencies of reinforcement).

Background to Behaviorism

Functional psychology represented the last remnant of the new psychology in America. Functional psychologists recognized the need for a socially useful psychology, but remained tied to the earlier philosophical tradition of James, Ladd,

and Hall; and their emphasis on the conscious and purposive adaptation of behavior harked back to pre-Darwinian teleological conceptions of psychological functions.

Although functional psychologists such as Angell and Carr acknowledged the value of structural psychology and the reliability of introspection, the experimental analysis of consciousness was neglected in practice at the University of Chicago, as it came to be in most places (outside of Cornell) in the early decades of the 20th century. Moreover, while earlier studies of animal behavior at Chicago (and Clark and Harvard) were designed to throw light on the comparative psychology of different species, animal psychology came to focus almost exclusively on the experimental study of animal learning, conceived of as a way to study the fundamental laws governing all animal and human behavior. In this fashion the advocacy of animal psychology by functional psychologists eventually came to undermine their own position, as animal psychologists developed explanations of animal learning that made no appeal to consciousness or cognition. As the rigor of the experimental study of animal learning came to displace the rigor of the experimental study of human consciousness as the paradigm of scientific objectivity, the methods of the animal laboratory were generalized to the study of human behavior.

Early Forms of Behaviorism The conceptual foundations of behaviorism were developed by the **new realist** philosophers, notably Edwin Bissell Holt (1873–1946) and Ralph Barton Perry (1876–1957) at Harvard and Edgar Arthur Singer (1873–1955) at the University of Pennsylvania (Mills, 1998). These philosophers maintained that consciousness and cognition are best explicated in terms of behavioral adjustments to the environment (Holt, 1914; Holt et al., 1910; Perry, 1904) to the point of the virtual equation of cognition and adaptive behavior in the work of Singer (1911, 1924) and Grace Mead Andrus de Laguna (1919).

Max Meyer advanced the earliest statement of the behaviorist position in *The Fundamental Laws of Human Learning* in 1911, followed by *The Psychology of the Other One* in 1921. Meyer, a former student of Stumpf's, accepted an appointment at the University of Missouri in 1900, where he remained for the rest of his career (Esper, 1966). He claimed that psychology could attain the status of a science of prediction and control only when the "facts and laws of introspective psychology have been correlated with—replaced by—facts of behavior and its laws" (1911, p. 241). Meyer also anticipated Watson's account of habit learning in terms of the recency and frequency of correlated stimuli and responses and his treatment of cognitive processes as motor responses. Meyer's work was largely neglected, possibly because his statement of behaviorism was tied to a very specific neurophysiological theory of stimulus-response learning. He also suffered the same type of academic fate as Watson: He was fired in 1930 for distributing questionnaire data on illicit sexual relations in his course on social psychology (O'Donnell, 1985).

Albert Paul Weiss's radically reductive version of behaviorism was also neglected (although Weiss, a modest and retiring man, made little attempt to publicize his views and died at an early age after an incapacitating illness). He trained as an undergraduate and postgraduate student at the University of Missouri, but spent the rest of his career at Ohio University. Weiss's comprehensive behaviorist approach accommodated all forms of behavior, from muscle twitches to socially embedded forms of thought and behavior (Weiss, 1917, 1918, 1925). He held that traditional problems of consciousness and cognition had been resolved or dissolved by the new realist philosophy (Mills, 1998) and defended the autonomy and pragmatic rationale of behaviorist psychology:

The success of behavior methods will not depend on how they treat the problem of consciousness; they will succeed or fail according as they do or do not further the general welfare of society.

—(1918, p. 637)

William McDougall: Purposive Behaviorism The Englishman William McDougall (1871–1938) advanced another early statement of behaviorism. In his *Introduction to Social Psychology* he claimed that

psychologists must cease to be content with the sterile and narrow conception of their science as the science of consciousness, and must boldly assert its claim to be the positive science of the mind in all its aspects and modes of functioning, or as I would prefer to say, the positive science of conduct and behavior.

—(1908, p. 15)

McDougall trained in medicine and physiology at the universities of Cambridge and London. He studied psychology with Müller at Göttingen and taught at the universities of London and Oxford before joining the philosophy faculty at Harvard (he accepted the position previously held by James and Münsterberg). McDougall developed a form of purposive behaviorism: He rejected introspective psychology but supported mentalistic explanations of behavior and maintained that many forms of purposive or goal-directed behavior are grounded in instinct. Although McDougall was well aware of the danger of “cheap and easy” explanations of behavior that merely postulate a corresponding instinct for every behavior to be explained (McDougall, 1908), his increasingly detailed theoretical inventory of instincts became the target of critics who depreciated the role of instinct in the explanation of human behavior (Dunlap, 1919; Kuo, 1921; Watson, 1924/1930).

McDougall was a prolific writer who made a significant contribution to the fledgling science of social psychology (1908, 1920), although his “group-mind” conception of social psychology was displaced by the individualist conception

developed by Floyd Allport (1890–1971) and his followers (Allport, 1924, 1933). McDougall's behaviorism is particularly interesting because although he was committed to the view that scientific psychology is concerned with the explanation of behavior, he did not share the optimistic environmentalism of Watson and most other behaviorists. McDougall was a committed hereditarian who advocated positive and negative eugenic programs in *Is America Safe for Democracy?* (1921). His position demonstrated that commitment to a science devoted to the prediction and control of behavior did not entail commitment to environmentalism, although in practice most American behaviorists were environmentalists.

McDougall's defense of a Lamarckian account of evolution during the period in which it was critically assessed and rejected by experimental biologists, along with his advocacy of spiritualist research, generated a storm of protest among psychologists, who called for his dismissal from the APA. He was never a popular figure, although he was the declared victor in a famous debate with Watson about the nature and prospects of behaviorism (Watson & McDougall, 1929). McDougall resigned from his position at Harvard in 1926 and moved to Duke University in 1927, where he remained until his death in 1938. He was lampooned in the press for his support of Lamarckianism and spiritualist research, but did not deserve the vilification he received from his colleagues when he died. In announcing McDougall's death at the APA meeting in 1938, Knight Dunlap claimed that McDougall had done a great service to psychology by dying (Smith, 1989).

Animal Psychology

McDougall held that a science of behavior should be grounded in a comparative psychology based upon "the observation of the behavior of men and animals of all varieties under all possible conditions of health and disease" (1908, p. 15), and Watson and later behaviorists made the experimental study of animal behavior the foundation of their psychologies. Animal psychology had been developed at major psychological centers such as Clark, Chicago, Columbia, Hopkins, and Harvard, but was in a rather precarious position at the time Watson issued his behaviorist "manifesto" in 1913. Although animal psychology had become more rigorously objective and experimental and had developed a coherent research program directed to the study of animal learning, only a handful of psychologists devoted themselves to the study of animal behavior in the first decade of the 20th century. When Watson suggested to his friend Robert M. Yerkes that those interested in animal behavior should get together at the APA meeting in 1909, he listed only five other psychologists and two biologists (O'Donnell, 1985). In the first decade of the 20th century, animal and comparative psychology accounted for only about 4 percent of the experimental output of American psychology laboratories (Bruner & Allport, 1940).

There were a variety of reasons for this. Animal psychology, like comparative psychology in the tradition of Darwin, Romanes, and Morgan, was criticized by traditional philosophers and structural psychologists as largely irrelevant to human psychology, and institutions that supported animal psychology were subject to public criticism and campaigns by anti-vivisectionists. Most college administrators were reluctant to provide funds for the care and maintenance of cats, dogs, and raccoons (which were smelly, dirty and noisy), in addition to the chronoscopes and reaction timers of the traditional psychology laboratory.

Most PhD students who did dissertation research on topics in animal psychology ended up in education, and most doctoral students in psychology considered animal psychology a doubtful and precarious career choice. Both Watson and Yerkes were advised to redirect their research focus to issues in education, as Edward L. Thorndike later did with great success at Columbia. Animal psychology was seen as peripheral to human psychology and was the first to have its budgets cut and faculty retrenched in times of financial depression, as happened in the years following the financial crisis of 1907. For example, Lawrence Wooster Cole, who conducted influential research on passive and delayed learning in raccoons at the University of Oklahoma, was fired in 1908, as was Eliot P. Frost at Yale in 1912 (O'Donnell, 1985).

The career of Robert Mearns Yerkes provides a usefully illustrative example. Yerkes came to Harvard as a graduate student in zoology, but was persuaded by the philosopher Josiah Royce (1855–1916) to combine his interests in zoology and psychology into comparative psychology. He transferred to philosophy in 1899 and completed his degree in 1902. Münsterberg's special pleading secured him an instructorship rather than the lowly teaching fellowship he was originally offered at Harvard. Yet despite a series of publications on animal instinct and learning and an appeal by Royce to the president, Yerkes remained an instructor for the next six years. Increasingly isolated (he was disinvited from philosophy department meetings in 1906) and personally humiliated by the promotion of a mediocre Harvard alumnus philosopher from a wealthy Boston family, Yerkes asked Münsterberg to once again plead his case with the president. Münsterberg secured his promotion to assistant professor on the explicit condition that Yerkes emphasize "the more educational aspects of psychology" (letter to President C. W. Eliot of Harvard, 1908, cited in O'Donnell, 1985).

When Yerkes continued to doggedly pursue his own animal research without even a nod to educational psychology, Münsterberg withdrew his support and threatened to shut down the program in comparative psychology. Yerkes's position improved only when he accepted a position at Boston Psychopathic Hospital in 1913, where he worked with all-too-human subjects. When Yerkes was offered the position of state psychologist prior to the First World War, Harvard doubled his salary and gave him a half-time teaching schedule (at full pay).

Yerkes was joint author of the first general introduction to Pavlov's work on classical conditioning (Yerkes & Morgulis, 1909). He published an *Introduction to Psychology* in 1911, and that same year he and Watson founded the *Journal of Animal Behavior*. Yerkes developed a lifelong interest in primates while working with orangutans at a primate laboratory in California during a two-year sabbatical from Harvard. He developed his talent for organization and committee work as head of the Army Testing Project during the First World War, and after the war he promoted the development of professional psychological research through his service on various committees of the National Research Council. He returned to academic life as a research professor at Yale in 1924, and in 1929 realized his lifetime ambition when he managed to secure funding to establish the Yale laboratories of primate biology in Jacksonville, Florida (now the Yerkes laboratories). That same year he published his major work *The Great Apes: A Study of Anthropoid Life* (1929).

The Albino Rat The perceived relevance of the experimental analysis of animal behavior for human psychology changed for the better as psychology became increasingly applied. Yet as it did so, animal psychologists abandoned the original conception of a genetic psychology based upon the comparative naturalistic observation of animal and human development and began to focus on the experimental study of the learning abilities of a limited number of animal species, most notably the domesticated albino variant of the Norwegian rat (*Rattus norvegicus var albinus*).

The Swiss psychiatrist Adolf Meyer (1866–1950), who became Massachusetts state pathologist at the Worcester Hospital for the Insane in 1896 (with an academic appointment at Clark University), introduced the albino rat to America. He established a breeding colony, although he did little research himself. However, he persuaded the developmental neurologist Henry Donaldson (1857–1938) of the rats' value in experimental neurology and supplied him with his first laboratory animals.

Donaldson had been a student of Hall's at Johns Hopkins in the 1880s. When Hall moved to Clark in 1889, he took Donaldson with him as an assistant professor of neurology. Donaldson was one of the many Clark faculty lured to the newly instituted University of Chicago in 1892, where he became head of the department of neurology. At Chicago Donaldson and his students began to use albino rats in their research, and he served as one of the advisors for John B. Watson's dissertation on neural development and learning in the albino rat (Watson, 1903). In 1906 Donaldson moved to the Wistar Institute, taking four pairs of rats with him. These formed the basis of the Institute's commercial breeding program, which became the main supplier of albino rats to neurology and animal psychology laboratories in America (Logan, 1999).

The albino rat had a number of advantages for research in neurology and psychology, including its slow rate of physiological, neural, and psychological

maturation, which made it an ideal animal for the study of development and learning. Although Meyer and Donaldson had originally conceived of the experimental study of rat behavior as an integral component of the comparative study of species differences and diversity, with the commercial production of the albino rat the experimental study of rat behavior came to be seen as the means of attaining maximum generality in animal research. American psychologists treated the albino rat as a generic animal model that could be used to represent developmental and learning processes common to all vertebrates, including humans (Logan, 1999). In this fashion the assumption of strong continuity between human and animal psychology and behavior that underpinned the standard behaviorist generalization from animal to human behavior was built into the “industrial standard” production model of the albino rat.

Criteria of the Psychic There was continued debate in the late 19th and early 20th centuries about the scientific legitimacy of theoretical attributions of consciousness and cognition to animals. Yerkes tried to resolve these doubts by developing objective criteria such as neurophysiological complexity and behavioral plasticity (Yerkes, 1905a). Like Romanes and Morgan before him, he acknowledged that questions about the scientific legitimacy of theoretical attributions of consciousness and cognition applied equally to humans and animals:

Human psychology stands or falls with comparative psychology. If the study of the mental life of lower animals is not legitimate, no more is the study of human consciousness.

—(1905b, p. 527)

Watson, in his capacity as review editor of animal psychology for the *Psychological Bulletin*, became increasingly dissatisfied with such attempts to develop “criteria of the psychic” (Yerkes, 1905a). He expressed his skepticism about the possibility of consistently applying such criteria in a paper he presented at the 1908 meeting of the Southern Society of Philosophy and Psychology at Johns Hopkins, titled “A Point of View in Comparative Psychology” (Watson, 1909), in which he argued that animal psychology should be restricted to the identification of observable stimulus-response sequences. Five years later, Watson extended this argument to human psychology when he advanced his behaviorist position in his Columbia lecture “Psychology as the Behaviorist Views It” (1913a). In promoting his position, Watson derived support from the work of Edward L. Thorndike in America and Ivan Pavlov in Russia, both of whom had taken up Morgan’s implicit challenge to develop accounts of animal behavior without reference to consciousness and cognition and had suggested that these accounts could be generalized to the explanation of human behavior.

Edward L. Thorndike: The Law of Effect

When Conwy Lloyd Morgan delivered the Lowell Lectures at Harvard in 1896, Edward Lee Thorndike may have been inspired to take up his challenge to explain animal behavior without reference to consciousness and cognition. Thorndike was at Harvard at the time and may very well have been in the audience, although he never claimed to have attended the lectures and there is no evidence that he did (Stam & Kalmanovitch, 1988). However, it is clear that Thorndike knew of Morgan's work, and he acknowledged that his own animal experiments were a development of Morgan's studies of learning in chickens. Morgan had trained chickens to discriminate different types of corn through what he had called "trial-and-error" learning (Morgan, 1894/1977), a characterization that Thorndike later appropriated.

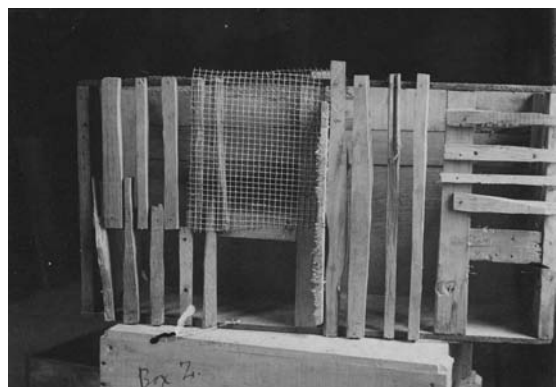
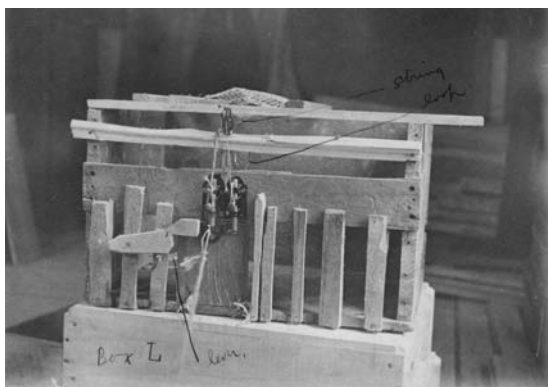
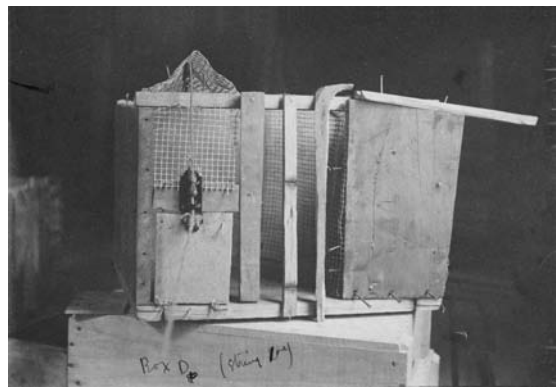
Thorndike was born in Williamsburg, Massachusetts, the son of a Methodist minister. From 1891 to 1895, he attended Wesleyan University, where he excelled academically and developed an interest in psychology after reading James's *Principles*. He attended Harvard, where he worked with and became friends with James, receiving his master's degree in 1897. Thorndike, who later confessed that he had had "no special interest in animals," worked with children on his original research project (Thorndike, 1936). However, the Harvard authorities refused to allow him to continue after Franz Boas conducted an anthropometric study that involved the loosening of children's clothes—creating a public outcry in Boston (O'Donnell, 1985). After consulting with James, Thorndike prepared to conduct a series of experiments on animal intelligence using chickens. When his landlady and the university refused to allow him space to conduct his experiments, James obliged by providing Thorndike with space in the basement of his house.

In 1897 Thorndike left Harvard for Columbia University. Cattell secured him a fellowship and encouraged him to continue his animal research. After difficulties with Thorndike's New York landlady, Cattell managed to secure some laboratory space at the university, where Thorndike conducted his famous studies of trial-and-error learning, based upon a series of experiments in which cats learned to escape from specially constructed "puzzle-boxes." His doctoral dissertation was accepted in 1898 and published that same year as *Animal Intelligence: An Experimental Study of the Associative Processes in Animals* (in the form of a monograph supplement in *Psychological Review*—it was republished in 1911 as an independent monograph). After a year teaching as an instructor in education at Case Western Reserve College for Women, Thorndike returned to New York as an instructor in genetic psychology at Teachers College, where he remained until his retirement in 1940.

The Law of Effect Thorndike's experimental studies of trial-and-error learning were based upon Morgan's reported observation of how his dog Toby had managed to escape from his back yard. Toby had learned to lift the latch on the gate after

repeated occasions in which originally accidental movements of the latch led to its opening and escape (Morgan, 1894/1977). Thorndike's cats were food-deprived and placed in slatted cages: They were required to open a latch (or series of latches) in order to escape and receive a food reward. Like Morgan's dog, the cats initially responded in a random fashion. They clawed and bit at the bars, sniffed around the cage, pushed their paws between the bars, and tried to squeeze between them. Eventually they hit upon the movement required to release the latch, which enabled them to escape and receive the food reward. On subsequent trials, the cats exhibited less and less random behavior until they learned the required behavior, which they would then produce whenever they were placed in the puzzle-box.

Thorndike measured the decreased time it took for animals to produce the required behavior and the decreased number of incorrect responses over the series of trials until learning was completed. His data indicated that the required response was learned incrementally, rather than through any spontaneous act of insight or reasoning. By varying the conditions, Thorndike also demonstrated



Thorndike's puzzle-boxes.

that learning was not a product of imitation, since the observation of successful responses by other cats did not decrease the time it took for a cat to learn the correct response.

Thorndike claimed that through this process of trial-and-error learning successful responses were “stamped in” and unsuccessful responses “stamped out”:

Gradually all the . . . nonsuccessful impulses will be stamped out and the particular impulse leading to the successful act will be stamped in by the resulting pleasure, until after many trials, the cat will, when put in a box, immediately claw the button or loop in a definite way.

—(1911, p. 36)

Thorndike characterized his account of this process as the **law of effect**:

Of several responses made to the same situation, those which are accompanied or closely followed by satisfaction to the animal will, other things being equal, be more firmly connected with the situation, so that, when it recurs, they will be more likely to recur; those which are accompanied or closely followed by discomfort to the animal will, other things being equal, have their connections with that situation weakened, so that, when it recurs, they will be less likely to occur. The greater the satisfaction or discomfort, the greater the strengthening or weakening of the bond.

—(1911, p. 244)

Thorndike supplemented the law of effect with what he called the **law of exercise**:

Any response to a situation will, other things being equal, be more strongly connected with the situation in proportion to the number of times it has been connected with that situation and to the average vigor and duration of the connections.

—(1911, p. 244)

Connectionism Thorndike’s theory of learning, which he characterized as **connectionism**, appealed to traditional principles of association based upon contiguity and repetition. However, he focused on associations between behavior and its consequences rather than associations between ideas (although the main principle of the law of effect had been recognized by Hartley, Spencer, and Bain). Indeed, part of the point of Thorndike’s experiments was to demonstrate that animal learning did not involve any form of insight, reasoning, or association of ideas. As Thorndike put it, “The effective part of the association [is] a direct bond between the situation and the impulse” (1911, p. 109). He believed this bond or connection to be automatic and unconscious.

Thorndike rather rashly concluded from these experiments that all animal and human behavior could be explained in terms of the laws of effect and exercise and the law of instinct:

The higher animals, including man, manifest no behavior beyond expectations from the laws of instinct, exercise and effect.

—(1911, p. 274)

For such conclusions he was roundly criticized, largely on the basis of the artificiality of his experimental studies.

Köhler (1917/1925) complained that Thorndike's cats were effectively forced into trial-and-error learning. According to Köhler, the cats were precluded from employing insight or reasoning because they could not see how the escape mechanism worked. Despite the fact that he agreed with Thorndike's conclusions, Morgan suggested that

The conditions of his experiments were perhaps not the most conducive to the discovery of rationality in animals if it exists. The sturdy and unconvinced advocate of reasoning (properly so-called) in animals may say that to place a starving kitten in the cramped confinement of one of Mr. Thorndike's box-cages, would be more likely to make a cat swear than to lead it to act rationally.

—(1898, p. 249)

The comparative psychologist Wesley Mills complained that the artificial and restricted nature of Thorndike's experimental setting precluded generalizations about normal animal and human behavior and famously remarked that one might as well "enclose a living man in a coffin, lower him, against his will, into the earth, and attempt to deduce normal psychology from his conduct" (Mills, 1899, p. 266).

Thorndike defended his theory and methods vigorously. He claimed that his manipulative experimental methods made his study scientifically superior to the previous "anecdotal tradition," of which he was openly contemptuous and dismissive (although he acknowledged Morgan's work and that of early experimental pioneers such as Spalding and Lubbock). Yet by the late 1920s he was forced to modify his theoretical position. He abandoned the law of exercise and the second half of the law of effect relating to the weakening of connections with "discomfort" (Thorndike, 1929).

Educational Psychology Although he was offered a position at Columbia (and later Harvard), Thorndike remained at Teachers College for the rest of his career, devoting most of his later years to educational psychology (Thorndike, 1903a, 1913–1914). He was a prolific author of academic articles and books, which prompted Titchener to

complain about Thorndike's tendency to "publish his lecture courses as soon as the lectures have been delivered" (1905, p. 522), as well as of texts and test instruments for educators (mainly on reading and arithmetic). The latter proved to be extremely lucrative, providing Thorndike with an income of around \$70,000 per annum in the 1920s (more than \$800,000 a year in today's dollars). He also ventured into the fields of industrial and social psychology, although with rather less success. Thorndike was the recipient of many awards and prizes, including honorary degrees. He was elected president of the APA in 1912, a member of the National Academy of Sciences in 1917, and president of the American Association for the Advancement of Science in 1934.

Although later behaviorists treated his research as exemplary, Thorndike himself was no behaviorist. He came to recognize the limitations of his account of learning with respect to distinctive human capacities such as language. He never denied the utility of introspection as a method of studying human consciousness or as a theoretical "window" into animal consciousness (Samelson, 1985). He was also a hereditarian who maintained that many human abilities and liabilities are innately determined. Thorndike published *Heredity, Correlation and Sex Differences in School Abilities* in 1903 and *Introduction to the Theory of Mental and Social Measurement* in 1904. He developed his own intelligence test, which was employed at Columbia and other educational institutions. He was also a committed eugenicist, who served with Henry H. Goddard and Robert M. Yerkes on the 1913 Committee on the Heredity of Feeble-mindedness, which recommended the compulsory sterilization of "mental defectives."

Later behaviorists treated Thorndike's research as exemplary because he did not appeal to consciousness or cognition in his connectionist account of trial-and-error learning. However, Thorndike did make reference to mental states such as "satisfaction," "annoyance," and "discomfort" in his statement of the law of effect, and he insisted that it was a virtue of his experimental method that it provided information about the feelings of animals. He also made the same inferences about the mental states of animals as had Romanes and Morgan:

For Thorndike, as for Morgan, detailed analysis of an animal's mental operations on the basis of objective inference was followed by descriptions of the animal's private experience on the basis of subjective inference.

—(Mackenzie, 1977, p. 70)

For example, Thorndike suggested that

One who has seen the phenomena so far described, who has watched the life of a cat or dog for a month or more under test conditions, gets, or fancies he gets, a fairly definite idea of what the intellectual life of a cat or dog feels like.

—(1911, pp. 123–124)

Ivan Pavlov: Classical Conditioning

Ivan Petrovich Pavlov (1849–1936) was born in the town of Ryazan, south of Moscow, the eldest son of a family of seven. His father was a parish priest, and Pavlov originally studied for the priesthood at Ryazan Ecclesiastical Seminary. He later developed an interest in natural science and enrolled at the University of St. Petersburg in 1870. He took a degree in natural science in 1879 and a medical degree at the Imperial Medical-Military Academy of St. Petersburg in 1883. After a few years in minor positions, he was appointed professor of pharmacology at the Imperial Medical-Military Academy in 1890 and professor of physiology in 1895. In 1891 he became director of the Institute of Experimental Medicine of St. Petersburg, the home of his research for the next 40 years. He was elected to the Russian Academy of Sciences in 1907.

Conditioned Reflexes During the first decade of his career, Pavlov focused his research on the physiological study of the digestive system, which won him the Nobel Prize in 1904. In his Nobel address, he mentioned the problem of “psychic secretions” that was to occupy him for the next three decades. Pavlov reported that his laboratory dogs salivated in reflexive reaction not only to food stimuli (in the form of meat powder), but also to associated stimuli, such as the sight or sound of the experimenter. Pavlov and his students investigated the determinants of these **conditioned responses** (originally characterized as conditional responses, since they were conditional upon the original reflex),¹ which he described in *Conditioned Reflexes* (1928).

Pavlov distinguished between an unconditioned or innate reflex and a conditioned or learned reflex. In the case of an unconditioned reflex, an unconditioned stimulus (US), such as food powder placed in a dog’s mouth, generates an unconditioned response (UR), such as salivation. In the case of a conditioned reflex, Pavlov and his students demonstrated that an originally neutral stimulus, such as the sound of a metronome, will function as a conditioned stimulus (CS) and elicit a conditioned response (CR), such as salivation, after repeated pairing of the originally neutral stimulus with the unconditioned stimulus. However, Pavlov never used a bell as a conditioned stimulus. The cultural myth that he did derives from the fact that the artist who transposed the illustration of his experimental set-up for an American translation of Pavlov’s work substituted a bell for the original metronome.

Pavlov and his students also identified the phenomena of **extinction** (the attenuation of a conditioned response when a conditioned stimulus is no longer

¹The Russian phrase *uslovnyi refleks* that Pavlov used can be translated as either “conditional reflex” or “conditioned reflex.” Although the former is closer to Pavlov’s original meaning (Todes, 1997), early English translations used the latter term, which has stuck.



Pavlov's dogs.

paired with an unconditioned stimulus), **spontaneous recovery** (the tendency of a conditioned stimulus to elicit a conditioned response after some time has passed since extinction), and **disinhibition** (the tendency of any strong stimulus to elicit a conditioned response after extinction). Natalia R. Shenger-Krestovnikova, one of Pavlov's students, induced an "experimental neurosis" in dogs, by pairing food stimuli with presentations of a circle but not an ellipse. As presentations of ellipses were made progressively more circular, the dogs manifested violent and erratic behavior. For Pavlov, this provided some support for his conviction that most neuroses and psychoses are a product of the "derangement" of inhibitory reflexes in the brain (Boakes, 1984).

Like Thorndike's explanation of trial-and-error learning, Pavlov's explanation of conditioned responses appealed to traditional principles of contiguity and repetition. He maintained that his experiments on conditioned reflexes provided "a solid foundation for associationist psychology" (Simon, 1957, p. 18). Pavlov tried (unsuccessfully) to accommodate trial-and-error learning by treating it as a form of classical conditioning after the Polish physicians Jerzy Konorski and Stefan Miller claimed that his explanation of conditioned salivatory responses did not generalize to motor responses. Konorski and Miller maintained that classical conditioning and instrumental conditioning (trial-and-error learning) are two distinct forms of learning (Boakes, 1984).

Also like Thorndike, Pavlov eschewed appeals to consciousness and cognition in the explanation of animal learning. He claimed that all animal and human behavior could be explained in terms of “the influence of external stimuli” without reference to any “fantastic internal world.” Pavlov’s psychology was reductively materialistic and atomistic. For Pavlov, the ultimate explanation of all innate and learned responses was neurophysiological in nature, and all complex conditioned reflexes were additive functions of elementary reflexes. Pavlov was highly critical of the work of the Gestalt psychologists and dismissed their explanations of animal behavior in terms of insight and creative problem solving.

Like Sechenov, Pavlov promoted a form of objective psychology based upon the concepts as well as the methods of experimental physiology, although he credited Thorndike as the first objective psychologist. He was critical of Wundt’s “new psychology,” with its appeal to autonomous psychical processes. He was not opposed to the study of consciousness per se, but dismissed introspection as a method of investigation and claimed that consciousness could only be studied via “scientifically based methods.” Pavlov also followed Sechenov in maintaining that all animal and human behavior is a product of innate or learned reflexes, both generative and inhibitory.

Pavlov was a dedicated experimentalist who poured all his financial resources into his laboratory. He and his wife lived frugally for many years in order to support it. She agreed to this so long as he agreed not to drink or smoke and to limit his social life. Pavlov was a perfectionist and disciplinarian in the laboratory and famously reprimanded a student who arrived late to the laboratory because he had been trying to avoiding street fighting during the Russian Revolution. By contrast, he was sentimental and impractical in his private life. He took one of his wife’s shoes with him on an overseas trip and was robbed of \$2,000 that protruded from his open briefcase while he waited for a train in Grand Central Station, New York. His wife often had to remind him to collect his wages. Unusually for his day, he accepted women and Jews as students in his laboratory (although he initially barred them), but would not tolerate explanations that appealed to consciousness—for such a sin a student could be fined (Boakes, 1984).

Pavlov lost his savings (including his Nobel Prize money) when the Bolsheviks liquidated assets after the revolution. His relations with Lenin and Stalin were often stormy, but he survived the revolution better than did most of his contemporaries (although at one point he contemplated transferring his laboratory to England or America). The Communist government generally valued and supported his work, and Lenin proclaimed him a hero of the revolution in 1921.

Bechterev and Motor Reflexes Vladimir M. Bechterev (1857–1927), who graduated from the Imperial Medical-Military Academy of St. Petersburg in 1878, also developed an objective psychology based upon reflexes. He received

his doctorate from the academy in 1881 and later studied with Wundt and du Bois-Reymond. Bechterev created the first Russian laboratory devoted to experimental psychology at the University of Kazan in 1885. He returned to the Imperial Medical-Military Academy in 1893 as professor of psychic and nervous diseases, but later left to found the Psychoneurological Institute with some of his colleagues. He published *Objective Psychology* in three volumes between 1907 and 1912 and *General Principles of Human Reflexology* in 1917. Bechterev maintained that the aim of objective psychology (or reflexology) was to “determine correlations between man and his environment, both physical, biological and, above all, social” (1917/1932, p. 33). Like Pavlov, he acknowledged that objective psychology had originally been developed by Americans such as Thorndike.

Bechterev was somewhat disparaging of Pavlov’s own work. He claimed that Pavlov’s “association reflexes” had been known for years. This was certainly true: They had been identified by Whytt in the 18th century and by Spencer and Bain in the 19th century (Rosenweig, 1959). Bechterev’s own research focused upon conditioned motor responses associated with motor reflexes, such as the patellar (knee-jerk) reflex, and he was critical of Pavlov’s “saliva method” (which focused upon conditioned physiological reflexes).

John B. Watson: Psychology as the Behaviorist Views It

John B. Watson was not the first to adopt the behaviorist position, but was undoubtedly its most forceful and successful advocate. He was born and raised in Greenville, South Carolina (the second of five children), the son of a drunken philanderer and a devoutly religious woman. His father abandoned the family in 1891 when Watson was 13, and Watson never forgave him for it (in later years his father sought him out, but Watson refused to see him). A troublemaker during his high school years, Watson was arrested for rioting and discharging a gun in public. Due to his own persuasive powers or his mother’s Baptist connections, he managed to get himself accepted by Furman College in 1894 (at age 16), from which he graduated with a master’s degree in 1899 (at the age of 21). At Furman he was taught philosophy and psychology by Gordon B. Moore, who had spent a recent sabbatical at the University of Chicago, and who introduced Watson to the work of Wundt, James, Titchener, and Angell. Watson supported himself through college by working in the chemistry laboratory and then taught for a year in a local Greenville school. He promised his mother that he would become a minister and avowed this as his vocation while attending Furman (Creelan, 1974). However, Watson abandoned any interest he may have had in a religious vocation after his mother’s death in 1900, when he was accepted by the University of Chicago for graduate study in psychology.

Arriving in Chicago in 1900 with \$50 in his pocket, Watson worked as a waiter, a janitor in the psychology laboratory, and a caretaker in the animal laboratory of the neurologist Henry H. Donaldson to support himself during his three years of graduate study. He studied experimental psychology with Angell and philosophy with Dewey, whom he claimed never to have understood. Watson studied neurology (as a minor) with Donaldson, who taught him basic research techniques, and biology and physiology with Jacques Loeb, the physiologist famous for his reductive account of animal psychology and behavior in terms of basic associative processes underlying reflexes and tropisms. Watson's dissertation project was a developmental study of the neurophysiological conditions of learning in albino rats, supervised by Angell and Donaldson.

Watson submitted his thesis on "Animal Education: The Psychical Development of the White Rat, Correlated With the Growth of Its Nervous System" in 1903, which gained him his doctorate at the age of 25. Watson was the first to receive a PhD in psychology at Chicago as well as the youngest person to receive a PhD at Chicago. He published his thesis as *Animal Education* that same year, and it was favorably reviewed in the main psychology journals. Watson was hired as an assistant and instructor at Chicago, where he taught courses on animal and human psychology. He used Titchener's laboratory manuals in his courses on experimental psychology, although he later confessed that he was uncomfortable working with human subjects (1936, p. 276). That same year he married Mary Ickes, one of his students, who bore him two children.

Watson set about developing the animal laboratory at the University of Chicago. He began a program of research with Harvey Carr on the role of sensation in maze learning in rats, concluding that kinesthetic (muscular) cues form the basis of learning (Carr & Watson, 1908; Watson, 1907). In 1907 Watson began field studies of the migratory instincts of noddy and sooty terns on the Tortugas Islands near Key West, which gives the lie to later complaints about his exclusive emphasis on laboratory experimentation. During this time he stumbled upon the form of learning later characterized as *imprinting* by Konrad Lorenz (1935), although he seems to have missed its significance. Watson noted,

The birds have formed a great attachment to me. They will follow me all around the room. It is becoming more and more difficult to keep them in any box.

—(1908, p. 240)

Watson later claimed that he came to form his behaviorist position during his years at Chicago, but got little encouragement from his colleagues, and that Angell warned him against trying to develop a psychology that excluded the study of consciousness (Watson, 1936). In 1907 Baldwin offered Watson an assistant professorship at Johns Hopkins, but Angell made a matching offer

that was enough to persuade Watson to remain at Chicago. The following year Baldwin came back with an offer that Watson could scarcely refuse and Angell did not even try to match: a full professorship and a \$1,000 increase in salary. Watson later claimed that he left Chicago with great reluctance and would have remained if Angell had made an offer of even an associate professorship (Watson, 1936). However he had become increasingly disenchanted with the lack of support for his animal research at Chicago and was well aware of the virtues of an academic position within the Eastern academic establishment (O'Donnell, 1985). He was also attracted to Hopkins by the presence of Herbert Spencer Jennings (1868–1947) in the department of biology. Watson considered Jennings one of the “big three” in animal psychology, along with himself and Yerkes.

Watson was thus elevated to a full professorship in psychology at Johns Hopkins at the age of 30. His position was further advanced when the Baltimore police found Baldwin in a “colored house of prostitution” in 1908. The university dismissed him when they discovered that despite this embarrassing episode, Baldwin had accepted a nomination to serve on the Baltimore School Board (Pauly, 1979). Baldwin handed over the chairmanship of the psychology department and the editorship of *Psychological Review* to Watson before leaving for South America. In the ensuing years Watson's career went from strength to strength. He was elected president of the APA and the Southern Society of Philosophy and Psychology in 1914.

Knight Dunlap (1875–1949), a student of Münsterberg's who came to Hopkins in 1906, encouraged Watson in his more radical views. He later claimed to have persuaded Watson of his behaviorist principles, and Watson acknowledged his influence. Dunlap's paper “The Case Against Introspection,” published in *Psychological Review* in 1912, anticipated many of Watson's distinctive positions, although, like McDougall, Dunlap rejected introspection but accepted mentalistic explanations of behavior. Karl S. Lashley (1890–1958), who later went on to have a distinguished career in neurophysiology, enrolled as a graduate student in zoology at Hopkins in 1912 and worked with Watson on a variety of projects, including his field study of noddy and sooty terns (Watson & Lashley, 1915). He was an early supporter of Watson's behaviorism (Lashley, 1923), although his later research undermined the neurophysiological assumptions of Watson's theory of learning (Lashley, 1929).

In 1913 Cattell invited Watson to deliver a series of lectures on his “new psychology” at Columbia University. The first of these lectures, “Psychology as a Behaviorist Views It,” was published in *Psychological Review* that year. It became known as the behaviorist “manifesto,” although Watson himself never represented it as such—it was first called a “manifesto” by Woodworth in 1931. This was followed by a fuller statement of his behaviorist position in *Behavior: An Introduction to Comparative Psychology* in 1914.

Watson's Behaviorism Watson initiated what was later called the “behaviorist revolution” by declaring that

Psychology as the behaviorist views it is a purely objective experimental branch of natural science. Its theoretical goal is the prediction and control of behavior. Introspection forms no part of its methods, nor is the scientific value of its data dependent upon the readiness with which they lend themselves to interpretation in terms of consciousness.

—(1913a, p. 158)

He maintained that psychology is a positivist science of prediction and control, based upon the description of behavior and its observable antecedents. Watson's advocacy of a behaviorist psychology was one of many early-20th-century appeals for an atheoretical or “behavioral” approach in the human and social sciences. Similar positivist positions were developed by Luther Lee Bernard (1919) in sociology, Charles E. Merriam (1921) in politics, and Wesley Claire Mitchell (1925) in economics (Mills, 1998). Aside from his rejection of introspection, Watson's basic positivist position was remarkably similar to Titchener's. Like Titchener, he advocated an experimental science based upon repeatable observations, but focused upon observations of behavior rather than the introspection of mental states.

Watson's behaviorism was grounded in his disenchantment with the debate about animal mind and the problems of introspective psychology. However, it was not mandated by them, and Watson's radical position went beyond anything licensed by Morgan's canon or the “imageless-thought” debate. His behaviorist program was not based upon any established contrast between the success of his own program of animal research and the failure of introspective psychology. Watson's own limited experimental research on maze learning in rats scarcely provided the foundation for a general psychology encompassing all forms of animal and human behavior. His behaviorism was in fact little more than a restatement of traditional empiricist and associationist principles transposed from the realm of privately introspectable mental states to publicly observable behavior. The familiar “principle of contiguity” grounded Watson's stimulus-response laws, and the “principle of frequency” served as his explanation of “habit formation.”

By 1913, when Watson delivered his Columbia lecture, most psychologists had already moved beyond the experimental study of consciousness, and the work of functional and applied psychologists had established observable behavior as a legitimate subject of scientific psychology. Cattell, who had invited Watson to speak at Columbia, claimed in his 1904 address at the World's Fair in St. Louis:

I am not convinced that psychology should be limited to the study of consciousness as such . . . the rather widespread notion that there is no psychology apart from introspection is refuted by the brute argument of accomplished fact. It seems to me that

most of the research work that has been done by me or in my laboratory is nearly as independent of introspection as work in physics or in zoology.

—(1904, pp. 179–180)

At the 1910 meeting of the APA, Angell had predicted that behavior would displace consciousness as the primary subject matter of scientific psychology, and at the 1912 meeting he cautioned that it was threatening to replace consciousness altogether (Angell, 1913). Psychology had already come to be defined in terms of behavior in textbooks such as Walter B. Pillsbury's (1872–1960) *Essentials of Psychology* (1911, p. 1–2) and McDougall's *The Science of Human Behavior* (1912).

However, although these psychologists rejected the introspective method, they still retained mentalistic explanations of behavior (McDougall, 1908) or treated observable behavior as the basis for inferences about human and animal consciousness and cognition (Pillsbury, 1911). Watson complained that Pillsbury “went back” on his definition and considered conventional topics such as mental imagery (which was not surprising, since Pillsbury was a former student of Titchener's). In contrast, Watson claimed that the time had come for psychology to develop as a “science of behavior” and “discard all reference to consciousness.” According to Watson, psychology should abandon the terms “consciousness, mental states, mind, content, introspectively verifiable, imagery, and the like” (1913a, p. 166). He claimed that a science of behavior based upon the public observation of behavior could and should be developed in place of a science of consciousness based upon private introspection—a scientific psychology whose principles would be developed “in terms of stimulus and response, in terms of habit formation, habit integrations and the like” (1913a, p. 167).

Watson gave three main reasons for abandoning references to consciousness and cognition and focusing upon behavior. The first was that such references are redundant in animal psychology. Watson had come to the conclusion that speculation about the minds of animals had no theoretical or practical value in animal psychology:

More than one student in behavior has attempted to frame criteria of the psychic—to devise a set of objective, structural and functional criteria which, when applied in the particular instance, will enable us to decide whether such and such responses are positively conscious, merely indicative of consciousness, or whether they are purely “physiological.” Such problems as these can no longer satisfy behavior men. It would be better to give up the province altogether and admit frankly that the study of the behavior of animals has no justification, than to admit that our search is of such a “will o’ the wisp” character. One can assume either the presence or the absence of consciousness anywhere in the phylogenetic scale without affecting the problems of behavior by one jot or one tittle; and without influencing in any way the mode of experimental attack upon them.

—(1913a, p. 161)

He generalized this conclusion that one can study animal behavior without reference to consciousness or cognition to the study of human behavior:

It is granted that the behavior of animals can be investigated without appeal to consciousness. . . . The position is taken here that the behavior of man and the behavior of animals must be considered on the same plane.

—(1913a, p. 176)

Watson's second reason was the failure of structural psychology. He cited the lack of inter- and intrasubject agreement with respect to the method of "systematic experimental introspection" practiced by Titchener and his followers, including the so-called imageless-thought debate and the lack of agreement about theoretical definitions of conscious states. These were fair criticisms of structural psychology, but were insufficient grounds for the rejection of all theoretical references to consciousness and cognition in animal and human psychology.

Watson's third reason was the practical irrelevance of a psychology of consciousness. He noted that "experimental pedagogy, the psychology of drugs, the psychology of advertising, legal psychology, the psychology of tests, and psychopathology" had all managed to thrive while distancing themselves from the psychology of consciousness (1913a, p. 113). He maintained that psychology would attain substantive practical benefits by adopting a behaviorist approach:

If psychology would follow the plan I suggest, the educator, the physician, the jurist and the business man could utilize our data in a practical way, as soon as we are able, experimentally, to obtain them.

—(1913a, p. 168)

However, Watson's appeal to applied psychology was somewhat disingenuous, since he did not himself engage in any form of applied psychology until after he left academia. He steadfastly resisted attempts to persuade him to move into "experimental pedagogy" and had little time for "the psychology of tests." During the First World War he was almost court-martialed from the army because of a negative report he wrote questioning the validity of the personnel selection tests he was charged with supervising (Cohen, 1979).

Watson represented his advocacy of behaviorism as a generalization from his developed position in animal psychology. However, it was as much a plea for the legitimacy of animal psychology, independent of any bearing it might have on human psychology. In defense of a discipline of doubtful professional status, Watson argued that established laws governing the behavior of animals, from the lowly amoebae upward, "have value in and of themselves without reference to the behavior of man" (1913a, p. 177).

It was also somewhat misleading for Watson to represent his argument for a behaviorist human psychology as a generalization from animal psychology. This was because he claimed that humans have unique cognitive capacities that are based upon language, which animals lack. Watson maintained that

it can readily be understood that the search for reasoning, imagery, etc., in animals must forever remain futile, since such processes are dependent upon language or upon a set of similarly functioning bodily habits put on after language habits.

—(1914, p. 334)

In this fashion Watson reprised the Cartesian account of the fundamental difference between humans and animals in terms of cognition and language, based upon his conception of cognition as a form of “inner speech” located in the larynx:

The fundamental difference between man and animal from our point of view lies in the fact that the human being can form [linguistic] habits in the throat.

—(1914, p. 299)

In denying cognitive and linguistic capacities to animals, Watson denied strong continuity between human and animal psychology and behavior—“the continuity theory of the Darwinians” (1914, p. 321). However, he affirmed strong continuity with respect to the explanatory principles of habit learning (based upon contiguity, recency, and frequency), which he believed accounted for all forms of animal and human behavior. Watson maintained this position without contradiction because he treated all forms of cognition (including language) as motor responses subject to the laws of habit formation.

Cognition as Motor Response Watson rejected theoretical references to consciousness and cognition as illegitimate in human psychology, but only insofar as they were interpreted and defined in terms of introspective analysis. He did not deny the existence of human consciousness or cognition *per se*. He identified thought with inner speech, specifically with movements of the larynx: “All natural thought goes on in terms of sensori-motor processes in the larynx” (1913a, p. 174). This identification led Watson to make the bold prediction that if someone “suddenly lost his laryngeal apparatus without any serious injury to the other bodily mechanisms,” he would suffer a “serious limitation” in his ability to think (1914, p. 327). When this was falsified by documented cases of persons who continued to think after removal of their larynx, he modified his position to include other muscles involved in speech production. Watson believed that a method could be developed for observing thought by correlating overt speech with movements of the larynx, to obtain “a record similar in character to that of the phonogram” (1913b, p. 424), although he admitted that the ability to “study reflective processes by such

methods” would be “about as far off as the day when we can tell by physicochemical methods the difference in the structure and arrangement of molecules between living protoplasm and inorganic substances” (1913a, p. 174).

Watson claimed that thoughts are usually unconscious, since the sensory-motor processes involved in speech production “rarely come into consciousness in any person who has not groped for imagery in the psychological laboratory” (1913a, p. 174). For Watson, the virtue of identifying thought with movements of the larynx and other muscles involved in speech production was that it made thought an *observable behavior* on a par with other observable behavior and thus subject to prediction and control. As he put it, if such “implicit behavior” could be “shown to consist of nothing but word movements,” then the “behavior of the human being as a whole is as open to objective observation and control as the behavior of the lowest organism” (1913b, p. 424).

What was significant about Watson’s account was not his specific physiological location of cognition and language, but his conception of cognition and language as *peripheral motor responses*. He denied that consciousness and cognition play any role in the explanation of human behavior because he maintained that consciousness and cognition are themselves behavioral responses rather than “inner causes” of behavior. His equation of thought with “inner speech” enabled him to conclude that “reflective processes” are “as mechanical as habit” (1913a, p. 174). Sechenov had also denied that consciousness and cognition are explanatory causes of behavior, but Watson’s position was more radical. Sechenov had maintained that thought is only the proximate cause of behavior, which is itself determined by external environmental stimuli. Watson denied that thought is even the proximate cause of behavior. He maintained that thought is a behavioral response to environmental stimuli that plays no role in the generation of other forms of behavior: He treated thought as an epiphenomenal by-product of the habit mechanisms underlying stimulus-response laws. Watson thus repudiated the central thesis of functional psychology that consciousness and cognition play an “active role in the world of adjustment” (1913a, p. 166).

Watson denied that cognitive states are “centrally initiated” causes of behavior and consequently rejected the ideomotor theory of behavior (1913a, p. 174). He claimed that thoughts are motor responses that can be “integrated into systems which respond in serial order (associative mechanisms)” (1913a, p. 174). Assuming this to be the case, Watson maintained that there is “no theoretical limitation of the behavior method” (1913a, p. 174).

Conversely, in his second Columbia lecture, “Image and Affection in Behavior” (1913b), and in *Behavior: An Introduction to Comparative Psychology* (1914), Watson acknowledged that if cognition was a “centrally initiated process” that played a causal role in the generation of behavior, this would undermine his conception of a behaviorist psychology grounded in laws describing observable stimulus-response

sequences. This was because Watson held that the brain and central nervous system are nothing more “than a mechanism for coordinating incoming and outgoing impulses” (1913b, p. 424). For Watson, the possibility of such “centrally initiated” cognitive processes represented the most serious “stumbling block” to the “free passage from structuralism to behaviorism.” As he admitted,

If thought goes on in terms of centrally aroused sensations, as is maintained by the majority of both structural and functional psychologists, we should have to admit that there is a serious limitation on the side of method in behaviorism.

—(1913b, p. 421)

Watson recognized that many psychologists and some “psychologically inclined neurologists” were tempted to locate autonomous cognitive processes in the higher regions of the cerebral cortex, but claimed that this form of neurophysiological dualism was little more than a rearguard action to maintain the traditional conception of an autonomous rational soul:

When the psychologist threw away the soul he compromised with his conscience by setting up a “mind” which was to remain always hidden and difficult of access. The transfer from periphery to cortex has been the incentive for driving psychology into vain and fruitless searches of the unknown and unknowable.

—(1913b, p. 424)

The Reception of Watson’s Behaviorism Watson’s Columbia lectures were well attended, but few rushed to embrace his theoretical position, and it took some time for his form of behaviorism to make inroads into American psychology (Samelson, 1981). Watson’s extreme position was rejected by friends and critics alike, such as Calkins, Cattell, McDougall, Washburn, Woodworth, and Yerkes. The strongest criticism came from Angell, who described Watson’s position as “scientifically unsound and philosophically essentially illiterate” (letter to Titchener, 1915, quoted in Larson & Sullivan, 1965, p. 342), although Angell refrained from making his views public out of respect for his former student. Titchener’s own response was surprisingly muted. However, Titchener refused to acknowledge behaviorism as a form of scientific psychology and insisted that it was merely a technology: “Watson is asking us, in effect, to exchange a science for a technology” (1914, p. 14).

This seems to have been because Titchener valued the experimental rigor of Watson’s animal studies but dismissed the theoretical rationale of behaviorism. As he put it in a letter to Watson in 1924, “It is quite true that the logic of the behaviorists is muddled. . . . The strength of the movement lies not in its fundamental logic but in its laboratory performance” (quoted in Larson & Sullivan,

1965, p. 348). Watson and Titchener formed a genuine friendship. Titchener was Watson's houseguest when Johns Hopkins hosted the 1910 meeting of the Experimentalists, and Watson talked of "introspectionists" such as Titchener as his "friendly enemies" (Watson, 1920, p. 97).

For most psychologists, and especially applied psychologists, the primary and enduring attraction of Watson's behaviorist vision was its technological promise of prediction and control (Samelson, 1981). Yet if the forms of applied psychology to which Watson appealed, such as "experimental pedagogy, the psychology of drugs, the psychology of advertising, legal psychology, the psychology of tests, and psychopathology" (1913a, p. 169) owed little or nothing to the experimental analysis of consciousness, they owed little or nothing to behaviorism either. Applied psychologists such as Scott, Witmer, and Wolfe endorsed Watson's appeal for a practically useful psychology, but eschewed his specific theories of behavioral learning.

Watson made some genuine converts, such as Walter Samuel Hunter (1889–1954), who published a number of works supportive of behaviorism (Hunter, 1922, 1925, 1928) and offered the first course on learning in American psychology (Mills, 1998). Within a decade of Watson's Columbia lecture, a number of texts appeared championing Watson's behaviorist position in general psychology (Dunlap, 1922; Smith & Guthrie, 1921), clinical psychology (Burnham, 1924), social psychology (Allport, 1924), and developmental psychology (Mateer, 1918). Yet many psychologists called themselves behaviorists merely as a convenient label for their own independent rejection of an experimental science of consciousness and commitment to applied psychological science. While the established academic "old guard" protested Watson's critical rhetoric, the "silent majority" of applied psychologists continued with their practical explorations in educational, industrial, and clinical psychology (O'Donnell, 1985).

Watson himself had supreme faith in the potential of behaviorism. He believed that a commitment to behaviorist principles would enable psychology to establish a system of laws such that "given the stimulus, psychology can predict what the response will be; or, on the other hand, given the response, it can specify the nature of the effective stimulus" (1919, p. 10).

Learning and Conditioning Watson claimed that the principles of recency and frequency were sufficient to explain habit formation. He believed that this enabled him to eliminate all references to associations made by an organism:

When the useless movements are eliminated the correct movements arise serially [on account of their greater frequency] without any chaining or linking in any material sense (bonds, connections, etc.). . . . Stated in other terms, we find no necessity for speaking of "associations."

—(1914, p. 260)

In later years, Watson abandoned this account of habit formation in favor of Pavlov's account of conditioned reflexes, beginning with his 1915 presidential address to the APA titled "The Place of the Conditioned Reflex in Psychology" (Watson, 1916).

Yerkes and Morgulis had introduced Pavlov's work to American psychologists in a paper they published in 1909. Watson also described it in some detail in *Behavior: An Introduction to Comparative Psychology* (1914), and he and his graduate students translated Bechterev's *Objective Psychology* (from the French version of 1913). However, Watson's sudden enthusiasm for conditioned reflexes was a little surprising, given the failure by Watson's colleagues to experimentally replicate either the Pavlovian salivatory reflex (Lashley, 1916) or the Bechterev finger-withdrawal reflex (Hamel, 1919). Watson based his presidential address upon a single study in which he generated conditioned heart rate and foot flexion reflexes in a dog through pairing of a signal with electric shock to its paw (Bolles, 1993). He had originally intended to present the results of experiments conducted in the summer and fall of 1915, in which he and Lashley had tried to record the "faint contractions of musculature" involved in subvocal thought. When this also proved a failure, Watson changed his topic to conditioned reflexes at the last minute.

Watson exploited the principles of classical conditioning to induce a conditioned fear of a white rat in an 11-month-old infant who came to be known as "little Albert." This experiment was based upon research on newborns that Watson had begun in 1916 at the Henry Phipps Psychiatric Clinic in Baltimore, directed by Adolf Meyer. Meyer, then professor of psychiatry at Johns Hopkins, approved of Watson's behaviorist approach and invited him to create a laboratory for the study of child development. As a result of this research, Watson came to believe that although there are a number of innate emotional responses, most are the product of conditioned learning. Working with Rosalie Rayner, one of his graduate students, Watson induced a conditioned fear of rats in Albert by repeatedly pairing the presentation of the white rat with a loud and frightening stimulus, created by smashing a metal rod against a pan lid (Watson & Rayner, 1920). Watson and Rayner intended to extinguish Albert's fear of rats, but he was removed from the institution before they were able to do so (Harris, 1979).

Life's Little Difficulties By 1919, with the publication of *Psychology From the Point of View of a Behaviorist*, Watson was at the pinnacle of his academic career, and behaviorism was beginning to have a significant impact. However, his academic career came to an abrupt end the following year. He was publicly humiliated as a result of his affair with his graduate student, Rosalie Rayner. While visiting Rayner's parents, Watson's wife discovered love letters between Watson and Rayner, in which he had pledged that "every cell I have is yours." The press pilloried Watson during the sensational divorce trial that followed, in which the judge castigated Watson as "an expert in misbehavior." He was fired from Johns Hopkins in 1920,



John B. Watson and Little Albert with white rat.

although not because of his affair or his divorce. Watson and Rayner had developed laboratory instruments to measure the physiology of the female orgasm. Meyer demanded that the university fire Watson as a “matter of principle,” and Watson was terminated on the grounds of the “moral delinquency” of his orgasm research (Rodgers, 2001).

This made it very difficult for Watson to secure another academic position, and he never did (although he did secure a part-time position as a lecturer at the New School of Social Research in New York). Titchener was one of the few psychologists willing to defend him and to provide him with a letter of reference. Undeterred by what he later called one of “life’s little difficulties,” Watson turned to a career in advertising, in which he achieved even greater success than he had in academia. He developed strategies for predicting and controlling consumer behavior that were far more effective than any of his earlier forays into animal and human psychology. Watson started with J. Walter Thompson, rising to a vice-presidency and salary of around \$70,000 (over \$800,000 today) by 1930, after which he transferred to William Esty & Co. from 1935 to 1945. He increased the sales of products such as Johnson’s Baby Powder, Pond’s Cold Cream, and Maxwell House Coffee and initiated marketing strategies such as demographic surveys and free samples. He invented the coffee break to sell coffee and introduced candy at supermarket checkouts to ambush mothers and their children.

Although he no longer held an academic position, Watson continued to publish articles and books, including *Behaviorism* in 1924. During the 1920s, mainstream psychology journals continued to publish discussions of his work, although references to it began to decline precipitously after 1935 (Todd, 1994). A Laura Spelman Foundation grant in 1923 enabled Watson to work with Mary Cover Jones on the elimination of conditioned fears (such as those induced in little Albert). Watson and Cover Jones eliminated a fear of rabbits from a child known as Peter (Jones, 1924) via “counterconditioning,” by teaching the child to relax in progressive approximations to the fear-inducing stimulus—a technique later characterized as “systematic desensitization” (Wolpe, 1958, 1973).

Watson’s Environmentalism In his last major work, *Behaviorism* (1924/1930), Watson virtually denied the existence of inherited instincts, traits, and abilities (although he had earlier granted the instinctual basis of many forms of animal and human behavior [Watson, 1914]) and proclaimed that, given the resources, a behaviorist psychology could be employed to create any type of human being that society desired:

Give me a dozen healthy infants, well-formed, and my own specified world to bring them up in and I’ll guarantee to take any one at random and train him to become any type of specialist I might select—a doctor, lawyer, artist, merchant-chief and yes, even into beggarman and thief, regardless of his talents, penchants, tendencies, abilities, vocations and race of his ancestors.

—(1924/1930, pp. 103–104)

Watson’s extreme environmentalism was not mandated by his positivist commitment to the description, prediction, and control of observable behavior as the goal of scientific psychology. However, it was a view shared by many other behaviorists, especially those critical of McDougall’s theory of instincts, such as Knight Dunlap (1919) and Luther Bernard (1921). Perhaps the most extreme environmental behaviorist was Zing-Yang Kuo (1898–1970), a student of Edward C. Tolman’s at the University of California at Berkeley, who wrote a series of influential critiques of the concept of instinct (1921, 1924, 1930). Kuo conducted a series of experiments in which kittens were raised with rats and birds, but failed to display supposedly instinctive aggressive and predatory behavior. Kuo returned to China in 1923, where he is said to have introduced behaviorism (Hothersall, 1995), although recent research has cast doubt upon his commitment to it (Blowers, 1998).

Last Years In 1928 Watson and Rosalie Rayner (now Rosalie Watson) published *The Psychological Care of Infant and Child*. This popular work, which was favorably reviewed by the philosopher Bertrand Russell (who particularly approved of

Watson's advocacy of sex education), sold over 100,000 copies in the first few months. They recommended an austere approach to child-raising, supposedly based upon behaviorist principles, in which overt love and affection were constrained:

Never hug and kiss them, never let them sit on your lap. If you must, kiss them once on the forehead when they say good night. Shake hands with them in the morning.

—(1928, p. 81)

Such practices were doubtfully effective with Watson's own children. His eldest son, James Watson, later described his father as "unresponsive, emotionally uncommunicative, unable to express and cope with any feelings or emotions of his own" (Hannush, 1987, p. 137). James attempted suicide, and his brother Billy, a psychiatrist and chronic alcoholic, committed suicide. Rosalie later expressed doubts about some of their parenting practices in an article in *Parents Magazine* titled "I Am the Mother of a Behaviorist's Sons" (R. R. Watson, 1930).

Aside from his academic publications and popular books, Watson also published regularly in fashionable magazines such as *Harper's*, *New Republic*, and *Cosmopolitan*. An articulate and persuasive publicist, Watson gave regular radio talks and newspaper interviews, and in later years he became somewhat of a celebrity. He built himself a mansion in Connecticut from the proceeds of his popular books and remuneration from his work in advertising. Watson was by all accounts a handsome and charming man and a snappy dresser who enjoyed the good life. He was also a hard drinker and incorrigible womanizer. Although McDougall officially won the debate with Watson known as "the battle of behaviorism" (Watson & McDougall, 1929), he wryly noted that Watson had secured all of the female votes.

While Watson surmounted many of "life's little difficulties," he never managed to cope with the death of his wife Rosalie in 1935 (from a fever contracted on an overseas trip). He had more or less abandoned psychology by then and in later years became somewhat of a recluse. He sold his mansion in Connecticut and drowned his sorrows in alcohol when he failed to consume them with his work in advertising.

Watson was awarded an APA citation in 1957 for his contribution to psychology, which stated that he had "initiated a revolution in psychological thought" (Karier, 1986, p. 148). At the age of 79, he traveled to the New York hotel where the convention ceremony was being held, but at the last moment could not bring himself to go in—his eldest son Billy went instead to acknowledge the honor (Buckley, 1989). Watson's legacy remained ambiguous for many psychologists, including neobehaviorists, who were reluctant to acknowledge him as an intellectual predecessor. It was hard to find someone willing to write Watson's obituary when he died the following year (according to Franz Samelson, cited in Mills, 1998).



On the farm: John B. Watson in retirement.

MENTAL TESTING, IMMIGRATION, AND STERILIZATION

Watson and later behaviorists extended their principles of learning based upon experimental studies of animal behavior to encompass all forms of animal and human behavior and treated them as the foundation of their technologies of prediction and control. This explanatory imperialism exposed them to later critiques of the scope of their theories by ethologists, animal psychologists, and cognitive psychologists. However, behaviorists were not the only ones who overreached themselves in the early decades of the 20th century. Other psychologists realized Wundt's and Titchener's fears in their attempt to transform psychology from a "science of trivialities" to a "science of human engineering" through programs of mental testing (Terman, 1924, p. 106).

Cattell's attempt to develop mental testing at Columbia based upon Galton's theory that sensory acuity is significantly correlated with intelligence came to grief when Wissler's studies indicated that there was no significant correlation between mental test scores and course grades (Wissler, 1901). However, mental testing did not die out with Cattell's anthropometric program. The British psychologist Charles Spearman (1863–1945), a former student of Wundt's, was highly

critical of Wissler's results and claimed that his own research demonstrated a strong correlation between measures of sensory acuity and intelligence (Spearman, 1904). He maintained that the various tests he employed were measures of a common capacity that he called "general intelligence" (g). A dedicated Galtonian, Spearman claimed that this capacity is determined by inheritance. Many American psychologists accepted this view and developed programs of mental testing based mainly upon measures of intelligence, but also upon measures of attitude, aptitude, development, mental health, and personality.

The Binet-Simon Intelligence Test

Alfred Binet (1857–1911), director of the laboratory of physiological psychology at the Sorbonne and editor of the journal *L'Année Psychologique*, developed a series of tests designed to identify retarded but educable children in the French elementary school system, which had expanded dramatically in the late 19th century when primary education became compulsory. Binet first described these tests, which included measures of association, sentence completion (derived from Ebbinghaus), memory, and moral judgment in *Experimental Studies of Intelligence* in 1903.

With his research assistant, Theodore Simon (1873–1961), Binet published a series of papers in *L'Année Psychologique* in 1905 describing a new scale for measuring child intelligence (Binet & Simon, 1905a, 1905b, 1905c). This scale, which included 30 items ranked in order of difficulty, was administered to large numbers of schoolchildren between 1905 and 1908. The scale was revised in 1908, with 54 tests arranged according to appropriate age levels between 3 and 13, ensuring that most children would test according to age (so that an average 10-year-old would test at age 10). The Binet-Simon scale (revised again in 1911, just before Binet's death) was a great success in Europe and was translated into a variety of languages. It provided an objective measure of general intelligence that was easy to administer, although Binet and Simon stressed the limitations of their test.

Binet and Simon employed the term "mental level" rather than the later and more popular "mental age" (Stern, 1914) to characterize the appropriate age levels. This was because they held that their scale was a useful instrument for identifying children in need of special remedial education and not a measure of a fixed level of intelligence. They devised special programs of education by which they hoped to raise the mental level of retarded children.

However, such caution was thrown to the winds when Henry H. Goddard and Lewis M. Terman (1877–1956) imported the Binet-Simon scale to America. Goddard, an energetic and enthusiastic graduate of Clark University, became director of the research laboratory for the study of feeble-mindedness at the Vineland Training School for Feeble-Minded Girls and Boys in New Jersey in 1906. He translated the 1908 Binet-Simon scale into English (with only minor revisions) and

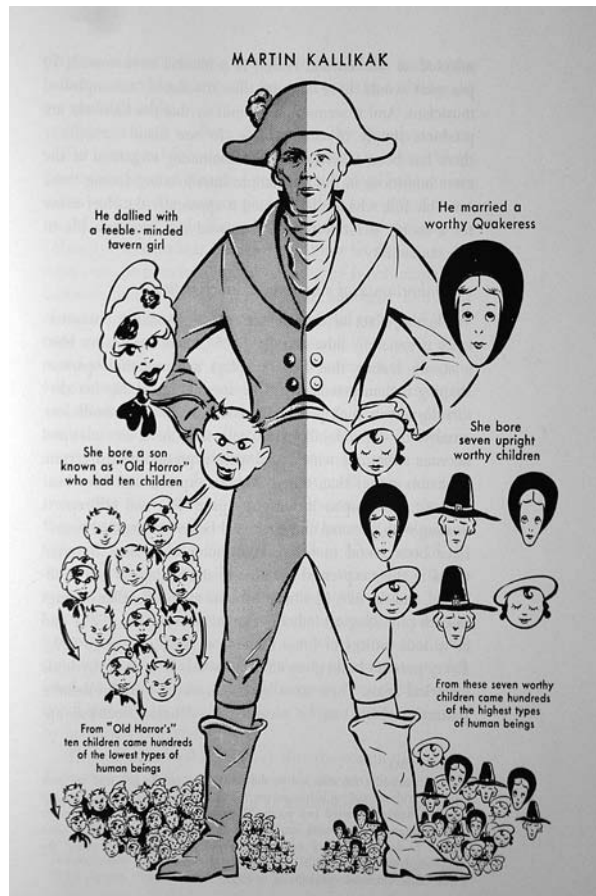
trained teachers at local schools to administer and score the test. He also translated the revised 1911 Binet-Simon scale (Goddard, 1911), which became the standard American intelligence test until Lewis Terman brought out his version in 1916.

Lewis M. Terman was a farm boy from Indiana who excelled at school and college and secured a fellowship at Clark University in 1903. He gained his PhD in 1905 with a bluntly titled dissertation on “Genius and Stupidity” (Terman, 1906). Terman moved to California for health reasons, and after a few years as principal of San Bernardino High School and as professor of pedagogy at Los Angeles State Normal College, he became professor of child study at Stanford University in 1910. There he remained for the rest of his career, serving as chair of the department of psychology for many years. Terman’s 1916 revision of the Binet-Simon scale was more than a translation: It included 36 additional items and was calibrated in relation to a substantial standardization sample drawn from California schools. The Stanford-Binet scale, as it came to be known, was designed to ensure that the average child at any age between 5 and 16 would test at that age. Thus the average 10-year-old would test at the “mental age” of 10, and the average “intelligence quotient” (IQ) for any age would be 100. Terman defined the **intelligence quotient** (IQ) as Stern’s “mental quotient” (the ratio of mental age to chronological age) multiplied by 100. The Stanford-Binet scale became the standard American intelligence test until it was itself revised in 1937.

Goddard and the Feeble-minded

Goddard and Terman did not follow Binet and Simon in treating their scales as merely useful instruments for identifying children in need of remedial education. Rather, they treated them as objective measures of genetically determined levels of intelligence, to be employed in interventionist programs of social engineering. Goddard’s study of a New England family, *The Kallikak Family: A Study in Feeble-mindedness* (1912), purported to demonstrate the inheritance of intelligence and feeble-mindedness. Goddard, who had read Galton and heard about Mendel’s work while visiting Germany, believed that intelligence and feeble-mindedness were inherited. He had learned from his experience at Vineland that the relatives of feeble-minded persons (individuals with an IQ of less than 70) were often feeble-minded themselves. When Goddard came across a girl of 22 who tested with a mental age of 9, whom he called Deborah Kallikak, he set about retracing her family tree.

According to Goddard and his researchers, Deborah’s family tree could be traced back to one Martin Kallikak, a Revolutionary War soldier. Martin Kallikak had come from good Quaker stock and had married a maidenly Quaker girl who bore him seven children. The descendants of this “good” side of the family tree went on to become upstanding citizens, who served as lawyers, doctors, judges, teachers, and landed gentry. However, Martin had also dallied with a feeble-minded



Representation of the “good” and “bad” descendants of Martin Kallikak.

serving wench while intoxicated. She had born him an illegitimate son, Martin Kallikak, Junior, who fathered seven children by his own marriage. The descendants of this “bad” side of the family tree went on to become immoral and sexually licentious horse thieves, prostitutes, brothel owners, and alcoholics—clear indicators of feeble-mindedness (the original Martin Kallikak’s own sexual licentiousness and intoxication was conveniently ignored).

Although Goddard called the study a “natural experiment,” there was no attempt to control for environmental or social differences, and his conclusion that feeble-mindedness is inherited was based upon the implausible assumption that it is determined by a recessive gene. Nonetheless, many accepted his conclusions, which were widely cited. Goddard’s study served as a paradigm for a host of similar (but equally flawed) studies of the inheritance of intelligence and feeble-mindedness and was seen as a scientifically reputable version of Richard Dugdale’s (1877) earlier study of the Jukes, an upstate New York family with an ancestry of degeneracy and criminality. The phrase “Jukes and Kallikaks” was later employed to describe problem families of dubious heritage (Zenderland, 1998).

Goddard also managed to persuade immigration officers at Ellis Island that he was able to identify the feeble-minded among the increasing numbers of eastern and southern Europeans of “poor stock” (in comparison to northern and western Europeans of “good stock”) then immigrating to the United States. The inspectors were particularly impressed by Goddard’s ability to identify the feeble-minded by sight and then have his identification “objectively” confirmed by their low scores on the Binet-Simon test (although the translators who administered the test complained that they would not have been able to answer questions about the New York Giants when they first came to America). As immigration inspectors began to make use of Goddard’s intelligence tests (and the intuitive judgments of Goddard’s Vineland staff), they refused entry to increasing numbers of immigrants on grounds of feeble-mindedness.



Intelligence testing at Ellis Island

The First World War and the Army Testing Project

The contribution of psychologists to the First World War enormously boosted the public profile of mental testing and American psychology. Titchener's Experimentalists were meeting at Harvard when America's entry to the war was announced on April 6, 1917. They convened a special session (from which Titchener excused himself) to discuss the proposed contribution of psychologists, which was chaired by Robert M. Yerkes, then president of the APA. Although he was primarily a comparative psychologist at Harvard, Yerkes also worked as a consulting psychologist at Boston State Psychopathic Hospital, where he developed the Yerkes-Bridges Point Scale of Intelligence (Yerkes, Bridges, & Hardwick, 1915).

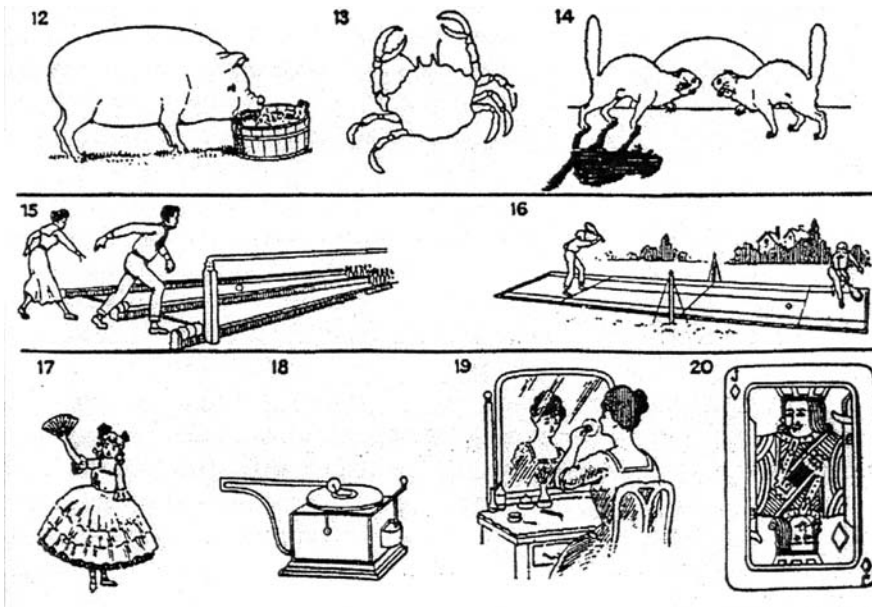
A born organizer, Yerkes lost no time in following the Experimentalists' recommendation that he visit Canada to study the role that Canadian psychologists had played in the war (as a British Dominion, Canada had entered the war with the Allies in 1914). A few days later, he met with Carl C. Brigham of the Canadian Military Hospitals Commission. At the meeting of the APA Council in Philadelphia later that month, Yerkes formed a committee, which included him, Cattell, Hall, Thorndike, and Watson, to explore the ways that psychologists could contribute to the war effort.

One proposal was to develop forms of psychological examination to facilitate the selection of officers and the discharge of feeble-minded recruits. Yerkes formed the Committee on Methods of Psychological Examining for Recruits, while Walter Dill Scott (who had a falling-out with Yerkes, whom Scott believed to be promoting his own interests) formed the Committee on the Classification of Personnel in the Army (CCPA). Yerkes's committee, which included Terman and Goddard, along with other educational, vocational, and medical testers, spent two weeks at Vineland creating intelligence tests for the army and running trials of these tests at local institutions and a few selected army bases. Recognizing that individual testing would be too time-consuming, they developed a number of group intelligence tests.

They submitted their proposal to test army recruits through the **National Research Council**, the national body founded in 1916 to administer government-funded natural science research projects. When the army approved what came to be known as the **Army Testing Project**, it created a division of psychology within the Sanitary Corps, which operated under the office of the Surgeon General (Scott's Committee on the Classification of Personnel operated under the office of the Adjutant General). The Army Testing Project eventually employed a team of over 400 army personnel who administered group intelligence tests to all army recruits: The Alpha test to literate soldiers, and the Beta (pictorial) test to those



Group intelligence testing of recruits in the First World War.



Army Beta Test: what's missing?

who could not read or could not read English. By the time the program ended in early 1919, they had tested close to 2 million soldiers.

Putting Psychology on the Map The practical utility of such mass testing was doubtful. Aside from the discharge of a small percentage of soldiers on grounds of mental incompetence (high levels of intelligence were not really necessary for the slaughter of trench warfare), the army accepted few of the recommendations of Yerkes's committee and discontinued the Army Testing Project after the war ended (Samelson, 1977). Nonetheless, many recognized that large-scale mental testing could have increased efficiency in the army and reduced costs (O'Donnell, 1985). The Army Testing Project, and the acknowledged success of Scott's committee on personnel classification, served to promote the public perception of the utility of mental testing and the public reputation of psychologists, with publications like the *New York Times* and *Harper's* magazine lauding their "achievement." As Cattell remarked, the Army Testing Project "put psychology on the map of the United States, extending in some cases beyond these limits into fairyland" (1922, p. 5).

Many psychologists who worked on the various army committees found the experience exciting and stimulating—and more rewarding than traditional academic life. Yerkes himself regretted that the war did not last longer! (O'Donnell, 1985). The emotional satisfaction of having their professional contributions recognized inspired and encouraged many psychologists to extend their practical war experience to the resolution of peacetime problems. Yerkes and Angell



Psychology Magazine, 1928.

were appointed to the National Research Council, which sponsored the development of the **National Intelligence Test** for children (National Research Council, 1920). This was financed by a Rockefeller grant of \$25 thousand and administered to over 7 million schoolchildren in the 1920s. Carl Brigham, the Canadian psychologist who later moved to Princeton, used the Army Alpha Test (which he had helped to develop) as the basis of the Scholastic Aptitude Test (SAT), which was first introduced as an admissions test for Princeton applicants.

Whatever the reality, the public believed that psychologists had developed a simple and effective means of measuring intelligence, and business and industry increased their demand for psychological testing services in order to promote their own efficiency and cost-cutting. The popularity of psychology increased dramatically in the 1920s, amidst the euphoria and general economic prosperity that followed victory in the First World War (Sokal, 1984). Promoters of the practical utility

of psychology wrote articles for magazines such as *Harper's*, *Colliers*, and *Atlantic Monthly*; and Joseph Jastrow contributed a syndicated daily newspaper column titled "Keeping Mentally Fit" (Jastrow, 1928, cited in Benjamin & Bryant, 1997). In his column "Exploring the Mind," the journalist Albert Wiggam touted the potential of psychology as an instrument of personal and social change:

Men and women never needed psychology as much as they need it today. You cannot achieve these things [effectiveness and happiness] in the fullest measure without the new knowledge of your own mind and the personality that the psychologists have given us.

—(1928, p. 13)

The 1920s saw the publication of the first wave of self-help books, with newspapers and magazines offering extensive psychological advice on a variety of issues, such as marriage, adolescence, depression, and child rearing. The 1920s

also saw the founding of the first popular psychology magazines (Benjamin & Bryant, 1997), such as *Psychology: Health, Happiness, Success* (founded 1923), *The Psychological Review of Reviews* (founded 1923), and *Industrial Psychology Monthly* (founded 1926). The public appetite for psychological knowledge and services provided new opportunities for educational, industrial, and clinical and counseling psychologists, but also for individuals with no psychological training who professed to psychological knowledge. Concerned with such developments, applied psychologists pressed the APA to develop a program of certification for psychologists, but with little success in the ensuing decades.

Immigration and Sterilization

Unfortunately, the new self-confidence of applied psychologists led them down a number of dangerous roads. One alarming finding of the Army Testing Project was that around half of the army recruits tested at or below the level of **moron**, defined by Goddard as between the mental age of 8 and 12 (with an IQ of between 50 and 70). Goddard claimed that the army test results demonstrated that “half the human race [is] little above the moron” (1919, p. 234). In his final report on the Army Testing Project, Yerkes concluded that “feeble-mindedness . . . is of much greater frequency than had previously been supposed” (Yerkes, 1921, p. 789). This generated a moral panic among psychologists and the public at large and accelerated two developments that had begun before the war, based upon assumptions about the inheritance of intelligence and feeble-mindedness: the demand for immigration quotas and sterilization programs.

A number of works painted bleak pictures of the consequences of allowing immigrants from eastern and southern Europe to contaminate the gene pool of the original Nordic (northern and western European) settlers. Carl Brigham argued in *A Study of American Intelligence* (1923, with a forward by Yerkes) that the average intelligence of recent immigrants was less than that of native-born Americans and that average intelligence had been decreasing significantly since 1900. Racist fears about the overbreeding of new immigrants (supposedly a sign of feeble-mindedness in itself) and claims about the intellectual and moral superiority of the Nordic race were developed in Madison Grant's *The Passing of the Great Race* (1916). McDougall's *Is America Safe for Democracy?* (1921) contained specific recommendations for programs of positive and negative eugenics to reverse the perceived downtrend in levels of intelligence. Goddard's own *Human Efficiency and Levels of Intelligence* (1920) expressed dark thoughts about the future of democracy and argued that the average mental age of other races, including blacks, was much lower than the average mental age of (male) whites. He recommended that intelligence tests should be employed to disenfranchise those of low intelligence, as well as to determine the suitability of job applicants.

A few psychologists urged caution and warned about the dangers of jumping to conclusions on the basis of limited data (Boring, 1923). Watson and the anthropologist Franz Boas vigorously challenged hereditarian assumptions about intelligence. Perhaps the most pointed critic was the journalist Walter Lippmann (1889–1974), who in a series of articles in the *New Republic* (which he founded) poured scorn upon the idea that psychologists had developed objective measures of intelligence. He dismissed the claim that the average intelligence of adult Americans was little above the level of a moron as equivalent to the absurd claim that the average intelligence of adults was below the average intelligence of adults (Lippmann, 1922). According to Lippmann, those who lamented the Army Testing Project estimate of the average mental age of 13 (Yerkes, 1921) were comparing this estimate with the average mental age of 16 estimated by the Stanford-Binet revision of 1916 (based upon a sample of white adults), which only demonstrated the inconsistency of the different estimates of average intelligence (a point echoed by Freeman, 1922). Lippmann identified the intelligence testing movement for what he believed it to be—an “engine of cruelty” based upon the “pretentiousness” of psychologists and the “abuse of the scientific method” (Passmore, 1978).

To little avail. The Army Testing Project also indicated a progressive decline in the intelligence of foreign-born soldiers when northern and western Europeans were compared to southern and eastern Europeans. Yerkes, in his introduction to Brigham’s *A Study of American Intelligence* (1923), echoed Galton’s and Pearson’s earlier concerns about British “national efficiency” by dramatizing “the menace of race deterioration” in America and “the evident relations of immigration to national progress and welfare” (Yerkes, 1923, pp. vii–viii, cited in Samelson, 1977, p. 278). Social and political concerns about the pollution of the “national stock” and Nordic gene pool by overbreeding hordes of Poles, Russians, Lithuanians, Jews, Irish, and other races led to the passing of the **National Origins Act** in 1924, which imposed quotas on each nationality based upon the 1890 census (before the post-1900 wave of eastern and southern European immigration).

Goddard and Terman served on the 1914 **Committee for the Heredity of Feeble-mindedness**, which also included Thorndike, Yerkes, Walter B. Cannon, and Alexander Graham Bell. The committee recommended a negative eugenics program: the sterilization of mental defectives, along with other “defective classes” such as criminals and the insane. The first state law licensing the sterilization of the feeble-minded was passed in Indiana in 1907, although it was eventually struck down by the state supreme court. However, other states followed the committee’s recommendation and introduced **sterilization laws** in the decades that followed. By the end of the 1920s close to 30 states had such laws on their books, and by 1930 around 12,000 people had been compulsorily sterilized.

This was the natural consequence of an age and scientific psychology “saturated with the development of a pragmatic theory of life” (Buchner, 1911, p. 3), and it represented the institutional realization of the pragmatist vision of James’s

Metaphysical Society at Harvard in the 1870s. Supreme Court Justice Oliver Wendell Holmes, one of the original members of the Metaphysical Society, wrote the majority decision in the case of *Buck vs. Bell* in 1927. The case challenged the right of the Commonwealth of Virginia to sterilize Carrie Buck, an 18-year-old inmate of the State Colony for Epileptics and Feeble-Minded. The commonwealth maintained that Carrie had a mental age of 9 years and that both her mother and 7-month-old daughter were feeble-minded (Zenderland, 1998). Writing in favor of compulsory sterilization, Holmes claimed,

It is better for all the world if instead of waiting to execute degenerate offspring for crime, or to let them starve for their imbecility, society can prevent those who are manifestly unfit from continuing their kind. . . . Three generations of imbeciles are enough.

—(1927, p. 207)

The dangers of such overreaching science were fully recognized only when the Nazis followed McDougall's (1921) recommendations for increasing the Nordic stock by introducing weekend retreats for blond and blue-eyed SS officers to breed with blonde and blue-eyed *fräulein* and created extermination camps as a logical extension of their own massive sterilization program for defective classes, including criminals, gypsies, and Jews. In 1930 Carl Brigham rejected his earlier racist views as "without foundation" and described his *Study of American Intelligence* (1923) as "one of the most pretentious of these comparative racial studies" (cited in Samelson, 1977, p. 278). (Brigham also later repudiated the use of the SAT for educational testing and opposed the formation of the Educational Testing Service.) By this time many of the more grandiose claims about the practical applications of scientific psychology to education, industry, and clinical practice were being questioned by a more skeptical public and more cautious government agencies, and the public image of psychology did not revive until after the Second World War. However, the immigration and sterilization laws remained in effect for a long time afterward. The Commonwealth of Virginia repealed the last sterilization law as late as 1981 (a formal apology to the victims of the law was issued 20 years later).

Yet the voices were not stilled for long. During the 1940s the British government instituted a national program of intelligence testing in schools (the "11-plus" test) that streamed students into academic and trade classes on the basis of their performance. The program was created by Cyril Burt (1833–1971), a committed hereditarian who held the chair in psychology at University College, London, founded by Galton. Burt was knighted in 1946 for his services to psychology and education, but in the mid-1970s serious doubts (including accusations of fabrication and fraud) were cast on the authenticity of his twin studies, which were frequently cited in support of hereditarian accounts of intelligence (Kamin, 1974; Samelson, 1992).

In the late 1960s Arthur Jensen's (1969) promotion of the claim that the (largely) genetic basis of intelligence renders educational interventions (such as Head Start) ineffective created a fierce debate and campus protests. In the 1990s Charles Murray and Richard J. Herrnstein reopened the debate with their claims about the genetic determination of intelligence and its correlation to socioeconomic status in *The Bell Curve* (Murray & Herrnstein, 1994).

THE STATUS OF APPLIED PSYCHOLOGY

While most of the American pioneers of scientific psychology did not share Wundt's and Titchener's antipathy to applied psychology and developed programs devoted to educational, industrial, and clinical psychology, they remained reluctant to grant the status of "scientific psychologist" to those whose psychological work was purely applied. The APA, which was originally founded by academic psychologists committed to experimental research, was also disinclined to accept purely applied psychologists, such as those working as clinical or consulting psychologists in hospitals or in private practice or those employed in testing and personnel selection for public companies. Part of the motivation for the creation of Titchener's Experimentalists (supported by applied psychologists such as Witmer, who had the original idea) was to develop a society of professional scientific psychologists, to distinguish themselves from unqualified charlatans who merely professed to have psychological knowledge.

In 1915 consulting psychologists petitioned the APA to create a program for the certification of psychologists, in order to protect the public and to establish their own professional status as experts. The APA leadership declined to act, citing the primary commitment of the association to the advancement of psychology as a science (Benjamin et al., 2003). Despite the lauded contributions of applied psychologists during the First World War, by the end of the war the 300-odd membership of the APA was restricted to academics with research credentials (a PhD, academic position in psychology, and publications in mainstream psychology journals).

Dissatisfied with this situation, Leta S. Hollingworth, J. E. Wallace Wallin, and Rudolf Pinter formed the American Association of Clinical Psychologists (AACP) in 1917. The aim of the association was to create professional standards for clinicians and mental testers and to attain legal recognition of clinical psychologists as authorities in the fields of psychopathology and mental retardation. Of the original 45 members of the AACP, some were affiliated with universities, while others worked in applied settings such as hospitals or child guidance clinics. The AACP sponsored a symposium on professional issues at the 1918 meeting of the APA, but the association was short-lived. Robert Yerkes, then president of the APA, managed to persuade the clinicians that their interests would be better served by becoming

a section of the APA. Yerkes was a member of both the APA and the AACP, but was deeply concerned about the dangers of a split between them.

In 1919 the AACP disbanded itself to become the Clinical Section of the APA, whose stated aim was to establish links between clinical psychology and other applied fields, to encourage research and publication, and to develop professional standards in clinical psychology (Napoli, 1981). A second APA section for consulting psychology was established in 1921, although requests for additional sections for industrial and educational psychology were denied (Routh & Reisman, 2003). The APA established a certification program of sorts in 1924, but merely issued certificates of professional acknowledgment that had no legal status. Few psychologists applied for them, and the program was abandoned in 1928 (Sokal, 1982). In 1925 the APA introduced a form of associate membership for applied psychologists who lacked academic credentials (such as employment in university departments and journal publications). This swelled the official membership of the APA but did little to advance the interests of applied psychologists, since the new associate membership did not carry voting rights.

Calls for a code of professional ethics and changes in graduate training to include applied experience and internships were largely ignored by the APA, although a committee of the Clinical Section of the APA was formed in 1931 to set standards for clinical training. Four years later the committee recommended that clinical psychologists should have a PhD and a year of practical experience (Report of Committee, 1935), although little was done to implement this recommendation (far less enforce it). Dissatisfied with the lack of support from the APA, the New York Association of Consulting Psychologists (founded in 1921), the largest of the state associations of applied psychologists, reorganized itself as the national Association of Consulting Psychologists (ACP) in 1930. The ACP pressed for legal standards and recognition and published a code of professional ethics in 1933. *The Journal of Consulting Psychology* was founded in 1937 as the official organ of the association.

The early 1930s saw the beginning of a development that later come to haunt the APA: The number of applied or “consulting” psychologists began to grow at a faster rate than the number of traditional academic or “scientific” psychologists. In 1938 the danger that Yerkes had foreseen came to pass. Dissatisfied with their influence within the APA, most of the members of the Clinical Division of the APA joined with members of the ACP to form the American Association for Applied Psychology (AAAP), which adopted the *Journal of Consulting Psychology* as its official organ. The Clinical Division of the APA and the ACP consequently disbanded themselves (although many AAAP members retained their APA membership). The AAAP originally comprised four sections, clinical, consulting, industrial, and educational, with their own officers, committees, and program at the annual meeting of the AAAP. Although the split was temporary, the tensions remained even after the AAAP disbanded when the APA was reformed and reconstituted in 1944 as a divisional federation (with a structure modeled upon the sectional organization of the AAAP).

DISCUSSION QUESTIONS

1. Did functional psychology owe anything to Darwin?
2. Woodworth dismissed the exclusive claims of the various schools of psychology and maintained that “every school is good, though no one is good enough.” Woodworth cited the schools of structural psychology, functional psychology, behaviorism, Gestalt psychology, and psychoanalysis. Can all these schools be held to be good together? Are they consistent?
3. Were “centrally initiated” cognitive states such a threat to Watson’s behaviorism as he supposed? Could he have accommodated them within his behaviorist system?
4. Although Binet and Simon stressed that their measures of child intelligence were merely useful instruments for identifying children who could benefit from remedial education, American translators and revisers of the Binet-Simon scale such as Goddard and Terman treated them as measures of genetically determined levels of intelligence. Why do you think this was the case? Do you think it had anything to do with their different national backgrounds?
5. Moral panic about the breeding of the feeble-minded fueled racist fears and led to immigration restrictions and sterilization programs. Are eugenics programs too dangerous to be implemented in the present day? While the dangers of negative eugenics are obvious, what are the dangers of positive eugenics (e.g., government support for the families of individuals who score high on intelligence tests)?

GLOSSARY

Army Testing Project The group intelligence testing of around 2 million army recruits during the First World War.

behaviorism 1. Form of psychology that treats observable behavior as the subject matter of scientific psychology. 2. Form of behaviorism developed by John B. Watson, who saw psychology as a positivist science restricted to the correlation of observable stimuli and responses, and rejected mentalistic explanations of behavior.

Buck vs. Bell (1927) Supreme Court case in which Carrie Buck’s appeal against compulsory sterilization was denied.

Committee for the Heredity of Feeble-Mindedness Committee established in 1914 whose members recommended the compulsory sterilization of mental defectives.

- conditioned response** An originally innate reflexive response that has come to function as a conditioned response through the repeated pairing of the originally eliciting stimulus with a neutral stimulus, which then comes to function as a conditioned stimulus.
- connectionism** Term employed by Thorndike to describe his theory of trial-and-error learning, in terms of automatic connections between situations and responses.
- disinhibition** The tendency of any strong stimulus to elicit a conditioned response after some time has passed since extinction.
- extinction** The attenuation of a conditioned response when a conditioned stimulus is no longer paired with an unconditioned stimulus.
- functional psychology** Form of psychology concerned with the functions of consciousness for adaptive behavior and associated with approaches to psychology developed at the University of Chicago and Columbia University.
- intelligence quotient** Measure introduced by Lewis Terman, defined as Stern's mental quotient (the ratio of mental age to chronological age) multiplied by 100.
- law of effect** According to Thorndike, responses that lead to satisfaction for an animal are more likely to be repeated, and responses that lead to discomfort are less likely to be repeated.
- law of exercise** According to Thorndike, the strength of a connection between a response and situation increases as a function of the vigor and duration of the connection.
- moron** As defined by Henry Goddard, an adult with a mental age between 8 and 12 (with an IQ between 50 and 70).
- National Origins Act** Congressional act of 1924 that restricted U.S. immigration to nationality quotas based upon the 1890 census.
- National Research Council** National body formed in 1916 to administer government-funded natural science research projects.
- National Intelligence Test** Group test of intelligence sponsored by the National Research Council and administered to over 7 million children in the 1920s.
- neobehaviorism** Form of behaviorist psychology originally developed by Edward C. Tolman and Clark L. Hull, which treated observable behavior as the subject matter of scientific psychology, but recognized the legitimacy of mentalistic explanations of behavior.
- new realism** Philosophical theory that held that consciousness and cognition are best explicated in terms of behavioral adjustments to the environment.

practice theory of reaction time Theory of Baldwin and Cattell that motor reaction times are often longer than sensory reaction times and that there are significant individual differences in both sensory and motor reaction times that are a product of differences in practice and attention.

radical behaviorism Form of behaviorism developed by B. F. Skinner and his followers, which treated observable behavior as the subject matter of scientific psychology, but rejected mentalistic explanations of behavior.

spontaneous recovery The tendency of a conditioned stimulus to elicit a conditioned response after extinction.

sterilization laws State laws passed in the early decades of the 20th century licensing the compulsory sterilization of the feeble-minded.

type theory of reaction time Theory of Wundt and Titchener that sensory reaction times are longer than motor reaction times.

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Neobehaviorism, Radical Behaviorism, and Problems of Behaviorism

WHILE MANY PSYCHOLOGISTS ACCEPTED WATSON'S RHETORIC OF prediction and control, few accepted the theoretical details of his behaviorist system, and in the 1930s and 1940s the neobehaviorism of Clark L. Hull (1884–1952) and Edward C. Tolman (1866–1959) superseded Watson's positivist brand of behaviorism. Hull and Tolman followed Watson in maintaining that scientific psychology should be directed to the explanation, prediction, and control of observable behavior rather than introspected mental states, and they rejected the form of structural psychology championed by Titchener and his followers. Yet in contrast to Watson, they recognized the legitimacy of theoretical explanations of observable behavior in terms of the internal states of organisms, including their mental states, on a par with theoretical explanations of the observable properties of physical elements in terms of their internal composition and structure (such as the explanation of the properties of carbon in terms of its molecular composition and structure).

The neobehaviorist attempt to more closely approximate the theoretical orientation of the natural sciences marked an advance over Watson's restriction of behaviorist psychology to the description of observable stimulus-response sequences. However, in their attempt to model behaviorist psychology upon the natural sciences, neobehaviorists did not look to the actual practice of natural sciences such as physics and chemistry, but adopted the equally restrictive logical positivist account of theory advanced by philosophers of science in the early decades of the 20th century. As Sigmund Koch put it,

In pursuit of these ends, psychology did not go directly to physics, but turned instead for its directives to middlemen. These were, for the most part, philosophers of science (especially logical positivists) and a number of physical scientists who had been codifying a synoptic view of the nature of science and who, by the early